# What is Money, Anyway?

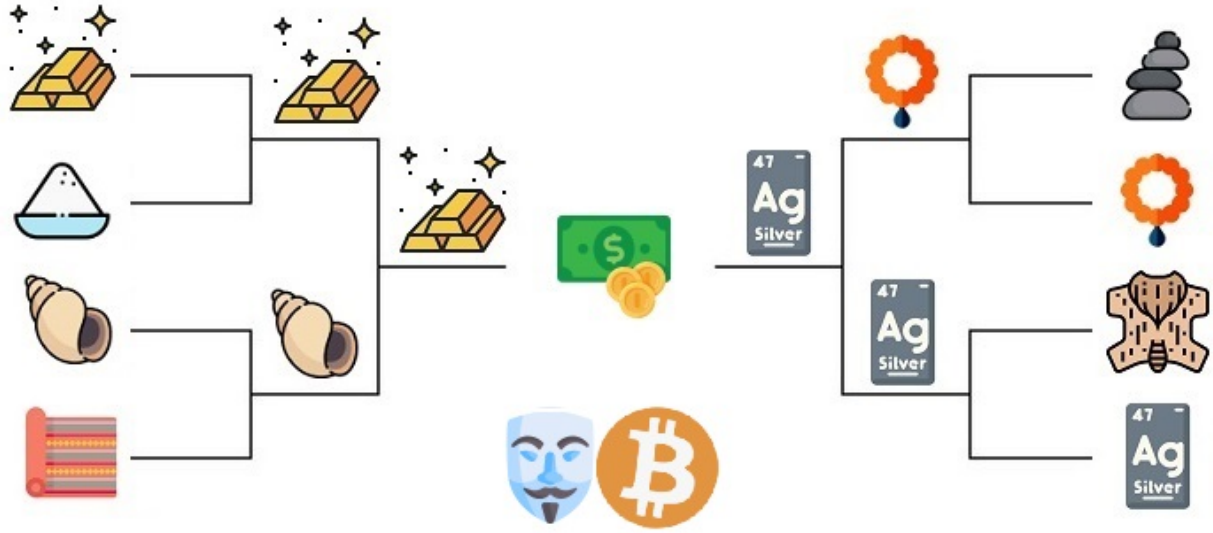



*Editor's Note:* This topic is now comprehensively covered in my book, *Broken Money*.

Money is a surprisingly complex subject.

People spend their lives seeking money, and in some ways it seems so straightforward, and yet what humanity has defined as money has changed significantly over the centuries.

How could something so simple and so universal, take so many different forms?




*Source of Icons: Flaticon*

It's an important question to ponder because we basically have four things we can do with our resources: consume, save, invest, or share.

**Consume:** When we consume, we meet our immediate needs and desires, including shelter, food, and entertainment.

**Save:** When we save, we store our resources in something that is safe, liquid, and portable, a.k.a. money. This serves as a low-risk battery of future resource consumption across time and space.

**Invest:** When we invest, we commit resources to a project that has a decent likelihood of multiplying our resources but also comes with a risk of losing them, by trying to provide some new value to ourselves or others. This serves

consumption potential compared to money. There are personal investments, like our own business or education, and there are external financial investments in companies or projects led by other people.

**Share:** When we share, or in other words give to charity and those in our community, we give some portion of our excess resources to those that we deem to be needing and deserving. In many ways, this can be considered a form of investment in the ongoing success and stability of our larger community, which is probably why we are wired to want to do it.

The majority of people in the world don't invest in financial assets; they are still on the consumption stage (basic necessities and daily entertainment) or the saving stage (money and home equity), either due to income constraints, consumption excesses, or because they live in part of the world that doesn't have well-developed capital markets. Many of them do, however, invest in expanding a self-owned business or in educating themselves and their children, meaning they invest in their personal lives, and they might share in their community as well, through religious institutions or secular initiatives.

Among the minority that do invest in financial assets, they are generally accustomed to the idea that investments change rapidly over time, and so they have to put a lot of thought into how they invest. They either figure out a strategy themselves and manage that, or they outsource that task to a specialist to do it for them to focus more on the skills that they earn the resources with in the first place.

However, depending on where they live in the world, people are not very accustomed to keeping track of the quality of money itself, or deciding which type of money to hold.

In developed countries in particular, people often just hold the currency of that country. In developing countries that tend to have a more recent and

what type of money they hold. They might try to minimize how much cash they hold and keep it in hard assets, or they might hold foreign currency, for example.

This article looks at the history of money, and examines this rather unusual period in time where we seem to be going through a gradual global transformation of what we define as money, comparable to the turning points of 1971-present (Petrodollar System), 1944-1971 (Bretton Woods System), the 1700s-1944 (Gold Standard System), and various commodity-money transition periods (pre-1700s). This type of occasion happens relatively rarely in history for any given society but has massive implications when it happens, so it's worth being aware of.

If we condense those stages to the basics, the world has gone through three phases: commodity money, gold standard (the final form of commodity money), and fiat currency.

A fourth phase, digital money, is on the horizon. This includes private digital assets (e.g. bitcoin and stablecoins) and public digital currencies (e.g. central bank digital currencies) that can change how we do banking, and what economic tools policymakers have in terms of fiscal and monetary policy. These assets can be thought of as digital versions of gold, commodities, or fiat currency, but they also have their own unique aspects.

This article walks through the history of monetary transitions from the lenses of a few different schools of thought (often at odds with each other), and then examines the current and near-term situation as it pertains to money and how we might go about investing in it. Start from the beginning or jump to the chapter you want:


- Chapter 1: Commodity Money
- Chapter 2: Fiat Currency

- Chapter 4: Final Thoughts

Some people whose work I've drawn from for this article, from the past and present, include Carl Menger, Warren Mosler, Friedrich Hayek, Satoshi Nakamoto, Adam Back, Saifdean Ammous, Vijay Boyapati, Stephanie Kelton, Ibn Battuta, Emil Sandstedt, Robert Breedlove, Ray Dalio, Alex Gladstein, Elizabeth Stark, Barry Eichengreen, Ross Stevens, Luke Gromen, Anita Posch, Jeff Booth, and Thomas Gresham.

# Commodity Money

> Money is not an invention of the state. It is not the product of a legislative act. Even the sanction of political authority is not necessary for its existence. Certain commodities came to be money quite naturally, as the result of economic relationships that were independent of the power of the state.
>
> -Carl Menger, 1840-1921

Humans in tiny groups don't need money; they can organize resources among themselves manually.

However, groups that reach the Dunbar number or larger usually start identifying and making use of some form of money, which gives them a more liquid, divisible, friction-minimized, and widely-accepted accounting unit for storing and exchanging value with people they don't know.

The more complex an economy becomes, the greater the number of possible combinations of barter you can have between different types of goods and services providers, so the economy starts requiring some standard unit of account, or money.

Specifically, the society begins requiring something divisible and universally acceptable. An apple farmer, for example, that needs some tools (a blacksmith), meat (a cattle rancher), repair work (a carpenter), and medicine for her children (a doctor), can't spend the time going around finding individuals that have what she needs, that also happen to want a ton of apples. Instead, she simply needs to be able to sell her highly seasonal apples for some unit, that she can use to save and buy all of those things with over time as she needs them.

Money, especially types of money that take work to produce, often seems arbitrary to outsiders of that culture. But that work ends up paying for itself many times over, because a standardized and credible medium of exchange and store of value makes all other economic transactions more efficient. The apple farmer doesn't need to find a specific doctor who wants to buy a ton of apples for his expensive services right now.

A number of economists from multiple economic schools have pondered and formulated this concept, but commodity money as a topic tends to come up the most often by those in the school of Austrian Economics, founded by Carl Menger in the 1800s.

In this way of thinking, money should be divisible, portable, durable, fungible, verifiable, and scarce. It also usually (but not always) has some utility in its own right. Different types of money have different "scores" along those metrics.

- Divisible means the money can be sub-divided into various sizes to take into account different sizes of purchases.
- Portable means the money is easy to move across distances, which means it has to pack a lot of value into a small weight.
- Durable means the money is easy to save across time; it does not rot or rust or break easily.

- Fungible means that individual units of the money don't differ significantly from each other, which allows for fast transactions.
- Verifiable means that the seller of the goods or services for the money can check that the money is what it really appears to be.
- Scarce means that the money supply does not change quickly, since a rapid change in supply would devalue existing units.
- Utility means that the money is intrinsically desirable in some way; it can be consumed or has aesthetic value, for example.

Summing those attributes together, money is the "most salable good" available in a society, meaning it's the good that is the most capable of being sold. Money is the good that is most universal, in the sense that people want it, or realize they can trade for it and then easily and reliably trade it for something else they do want.

Other definitions consider money to be "that which extinguishes debt", but debt is generally denominated in units of whatever money is defined to be at the time the debt was issued. In other words, debt is typically denominated in units of the most salable good, rather than the most salable good being defined as what debt is denominated in. Indeed, however, part of the ongoing network effect of what sustains a fiat currency system is the large amount of debt in the economy that creates sustained ongoing demand for those currency units to service those debts.

Back in 1912, Mr. J.P. Morgan testified before Congress and is quoted as having said the famous line:

> " Gold is money. Everything else is credit.

In other words, although their terms often overlap, currency and money can be thought of as two different things for the purpose of discussion.

We can define currency as a liability of an institution, typically either a commercial bank or a central bank, that is used as a medium of exchange and unit of account. Physical paper dollars are a formal liability of the US Federal Reserve, for example, while consumer bank deposits are a formal liability of that particular commercial bank (which in turn hold their reserves at the Federal Reserve, and those are liabilities of the Federal Reserve as well).

In contrast to currency, we can define money as a liquid and fungible asset that is not also a liability. It's something intrinsic, like gold. It's recognized as a highly salable good in and of itself. In some eras, money was held by banks as a reserve asset in order to support the currency that they issue as liabilities. Unlike a dollar, which is an asset to you but a liability of some other entity, you can hold gold which is an asset to you and a liability to nobody else.

Under gold standard systems, currency represented a claim for money. The bank would pay the bearer on demand if they came to redeem their banknote paper currency for its pegged amount of gold.

Scarcity is often what determines the winner between two competing commodity monies. However, it's not just about how rare the asset is. A good concept to be familiar with here is the stock-to-flow ratio, which measures how much supply there currently exists in the region or world (the stock) divided by how much new supply can be produced in a year (the flow).

For example, gold miners historically add about 1.5% new gold to the estimated existing above-ground gold supply each year, and the vast majority of gold does not get consumed; it gets re-melted and stored in various shapes and places.

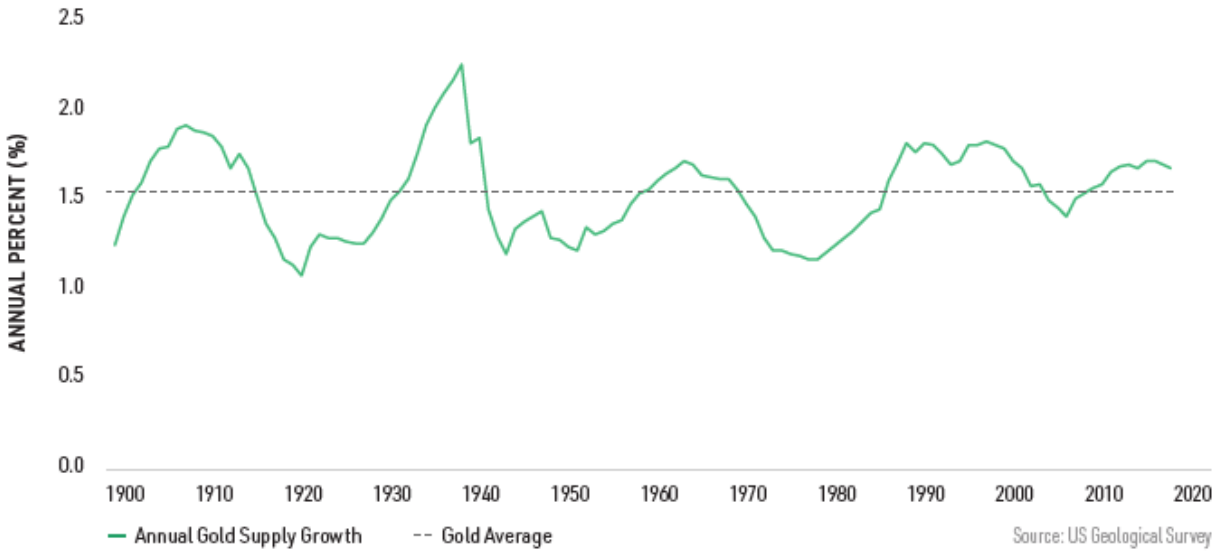



*Chart Source: NYDIG*

This gives gold a stock to flow ratio of 100/1.5 = 67 on average, which is the highest stock-to-flow ratio of any commodity. The world collectively owns 67 years worth of average annual production, based on World Gold Council estimates. Let's call it about 60 or 70 since this isn't exact.

If a money (the most salable good) is easy to create more of, then any rational economic actor would just go out and create more money for herself, diluting the whole supply of it. If an asset has a monetary premium on top of its pure utility value, then it's strongly incentivizing market participants to try to make more of it, and so only the forms of money that are the most resistant to debasement can withstand this challenge.

On the other hand, if a commodity is so rare that barely anyone has it, then it may be extremely valuable if it has utility, but it has little useful role as money. It's not liquid and widely-held, and so the frictional costs of buying and selling it are higher. Certain atomic elements like rhodium for example are rarer than gold, but have low stock-to-flow ratios because they are consumed by industry as quickly as they are mined. A rhodium coin or bar

So, a long-lasting high stock-to-flow ratio tends to be the best way to measure scarcity for something to be considered money, along with the other attributes on the list above, rather than absolute rarity. A commodity with a high stock-to-flow ratio is hard to produce, and yet a lot of it has already *been* produced and is widely distributed and held, because it either isn't rapidly consumed or isn't consumed at all. That's a relatively uncommon set of attributes.

Throughout history various stones, beads, feathers, shells, salt, furs, fabrics, sugar, coconuts, livestock, copper, silver, gold, and other things have served as money. They each have different scores for the various attributes of money, and tend to have certain strengths and weaknesses.

Salt for example is divisible, durable, verifiable, fungible, and has important utility, but is not very valuable per unit of weight and not very rare, so doesn't score very well for portability and scarcity.

Gold is the best among just about every attribute, and is the commodity with by far the highest stock-to-flow ratio. The one weakness it has compared to other commodities is that it's not very divisible. Even a small gold coin is more valuable than most purchases, and is worth as much as most people make in a week of labor. It's the king of commodities.

For a large portion of human history, silver has actually been the winner in terms of usage. It has the second-best score after gold across the board for most attributes, and the second highest stock-to-flow ratio, but beats gold in terms of divisibility, since small silver coins can be used for daily transactions. It's the queen of commodities. And in chess, the king may be the most important piece, but the queen is the most useful piece.

In other words, gold was often held by the wealthy as a long-term store (and display) of value, and as a medium of exchange for very large purchases, while silver was the more tactical money, used as a medium of exchange and store

regions of the world for that reason until relatively recently, despite the challenges that come with that.

The scarcity of some of the other commodities have more specific weaknesses as it relates to technology. Here are two examples:

**Rai Stones**

Inhabitants of a south-Pacific island called Yap used enormous stones as money. These “rai stones” or “fei stones” as they were called were circular discs of stone with a hole in the center, and came in various sizes, ranging from a few inches in diameter to over ten feet in diameter. Many of them were at least a couple feet across, and thus weighed hundreds of pounds. The biggest were over ten feet across and weighed several thousands of pounds.

Interestingly, I’ve seen this example used by both an Austrian economist (Saifedean Ammous) and an MMT economist (Warren Mosler). The reason that’s interesting is because those two schools of thought have very different conceptions of what money is.

Anyway, what made these stones unique was that they were made from a special type of limestone that was not found in abundance on the island. Yap islanders would travel 250 miles to a neighboring island called Palau to quarry the limestone and bring it back to Yap.

They would send a team of many people to that far island, quarry the rock in giant slabs, and bring it back on wooden boats. Imagine bringing a multi-thousand pound stone across 250 miles of open ocean on a wooden boat. People died in this process over the years.

Once made into rai stones on Yap, the big ones wouldn’t move. This is a small island, and all of the stones were catalogued by oral tradition. An owner could

moving the stone, this would take the form of announcing to the community that this other person owned the stone now.

In that sense, rai stones were a ledger system, not that different than our current monetary system. The ledger keeps track of who owns what, and this particular ledger happened to be orally distributed, which of course can only work in a small geography.

By the time this was documented by Europeans, there were thousands of rai stones on Yap, representing centuries of quarrying, transporting, and making them. Rai stones thus had a high stock-to-flow ratio, which is a main reason for why they could be used as money.

In the late 1800s, an Irishman named David O'Keefe came across the island and figured this out. And, with his better technology, he could easily quarry stone from Palau and bring it to Yap to make rai stones, and thus could become the richest man on the island, able to get locals to work for him and trade him various goods.

> "As the Irishman got to know Yap better, he realized that there was one commodity, and only one, that the local people coveted—the "stone money" for which the island was renowned and that was used in almost all high-value transactions on Yap. These coins were quarried from aragonite, a special sort of limestone that glistens in the light and was valuable because it was not found on the island. O'Keefe's genius was to recognize that, by importing the stones for his new friends, he could exchange them for labor on Yap's coconut plantations. The Yapese were not much interested in sweating for the trader's trinkets that were common currency elsewhere in the Pacific (nor should they have been, a visitor conceded, when "all food, drink and clothing is readily available, so there is no barter and no debt") , but they would work like demons for stone money

-Smithsonian Magazine, "David O'Keefe: the King of Hard Currency"

In essence, better technology eventually broke the stock-to-flow ratio of rai stones by dramatically increasing the flow. Foreigners with more advanced technology could bring any number of them to the island, become the wealthiest people on the island, and therefore increase the supply and reduce the value of the stones over time.

However, locals were smart too, and they eventually mitigated that process. They began to assign more value to older stones (ones that were verifiably quarried by hand decades or centuries ago), because they exclude the new abundant stones by definition and thus maintain their scarcity. Nonetheless, the writing was on the wall; this wasn't a great system anymore.

Things then took a darker turn. As described in that Smithsonian piece:

> "With O'Keefe dead and the Germans thoroughly entrenched, things began to go badly for the Yapese after 1901. The new rulers conscripted the islanders to dig a canal across the archipelago, and, when the Yapese proved unwilling, began commandeering their stone money, defacing the coins with black painted crosses and telling their subjects that they could only be redeemed through labor. Worst of all, the Germans introduced a law forbidding the Yapese from traveling more than 200 miles from their island. This put an immediate halt to the quarrying of *fei* , though the currency continued to be used even after the islands were seized by the Japanese, and then occupied by the United States in 1945.

Many of the stones were taken and used as makeshift anchors or building materials during World War II by the Japanese, reducing the number of

Rai stones were a notable form of money while they lasted because they had no utility. They were a way to display and record wealth, and little else. In essence, it was one of the earliest versions of a public ledger, since the stones didn't move and only oral records (or later, physical marks by Germans) dictated who owned them.

### African Beads

As another example, trade beads were used in parts of west Africa as money for many centuries, stretching back at least to the 1300s and prior as documented by ancient travelers at the time, as recorded by Emil Sandstedt. Various rare materials could be used, such as coral, amber, and glass. Venetian glass beads gradually made their way across the Sahara over time as well.

To quote Ibn Battatu, from his travels in the 14th century (from Sandstedt's article):

> " A traveler in this country carries no provisions, whether plain food or seasonings, and neither gold nor silver. He takes nothing but pieces of salt and glass ornaments, which the people call beads, and some aromatic goods.

These were pastoral societies, often on the move, and the ability to wear your money in the form of strands of beautiful beads was useful. These beads maintained a high stock-to-flow ratio because they were kept and traded as money, while being hard to produce with their level of technology.

Eventually, Europeans began traveling and accessing west Africa more frequently, noticed this usage of trade beads, and exploited them. Europeans had glass-making technology, and could produce beautiful beads with modest

Due to this technological asymmetry, they devalued these glass beads by increasing their supply throughout west Africa, and extracted a lot of value from those societies in the process. Locals kept trading scarce local "goods", ranging from important commodities to invaluable human lives, for glass beads that had far more abundance than they realized. As a result, they traded away their real valuables for fake valuables. Picking the wrong type of money can have dire consequences.

It wasn't as easy as one might suspect for the Europeans to accomplish, however, because the Africans' preferences for certain types of beads would change over time, and different tribes had different preferences. This seemed to be similar to the rai stones, where once new supplies of rai stones started coming in faster due to European technology, the people of Yap began wisely valuing old ones more than new ones. Essentially, the west African tastes seemed to change base on aesthetics/fashion and on scarcity. This, however, also gave that form of money a low score for fungibility, which reduced its reliability as money even for the pastoral west Africans who were using them.

Like rai stones ultimately, trade beads couldn't maintain their high stock-to-flow ratio in the face of technological progress, and therefore eventually were displaced as money.

**Japanese Invasion Money**

Although it's not a commodity money, the Japanese Empire used the same tactic on southeast Asians as the Europeans did on Africans.

During World War II, when the Japanese Empire invaded regions throughout Asia, they would confiscate hard currency from the locals and issue their own paper currency in its place, which is referred to as "invasion money". These conquered peoples would be forced to save and use a currency that had no backing and ultimately lost all of its value over time, and this was a way for

Japan to extract their savings while maintaining a temporary unit of account in those regions.

To a less extreme extent, this is what happens throughout many developing countries today; people constantly save in their local currency that, every generation or so, gets dramatically debased.

**Other Types of Commodity Money**

Emil Sandstedt's book, Money Dethroned: A Historical Journey, catalogs the various types of money used over the past thousand years or so. The book often references the writings of Ibn Battuta, the 14th century Moroccan explorer across multiple continents, who may have been the furthest traveler of pre-modern times.

Central Asians at the time of Battuta, as a nomadic culture, used livestock as money. The unit of account was a sheep, and larger types of livestock would be worth a certain multiple of sheep. As they settled into towns, however, the storage costs of livestock became too high. They eat a lot, they need space, and they're messy.

Russians had a history of using furs as a monetary good. There are even referenced instances of using a bank-like entity that would hold furs and issue paper claims against them. Parts of the American frontier later turned to furs as money for brief periods of time as well.

Seashells were used by a few different regions as money, and in some sense were like gold and beads in the sense that they were for both money and fashion.

In addition to beads, certain regions in Africa used fine fabric as money. Sometimes it wasn't even cut into usable shapes or meant to ever be worn; it

would be held and exchanged purely for its monetary value as a salable good that could be stored for quite a while.

Another great example is the idea of using blocks of high-quality Parmesan cheese as bank collateral. Since Parmesan cheese requires 18-36 months to mature, and is relatively expensive per unit of weight in block form, niche banks in Italy are able to accept it as collateral, as a form of attractive commodity money:

> MONTECAVOLO, ITALY (Bloomberg News) — The vaults of the regional bank Credito Emiliano hold a pungent gold prized by gourmands around the world — 17,000 tons of parmesan cheese.
>
> The bank accepts parmesan as collateral for loans, helping it to keep financing cheese makers in northern Italy even during the worst recession since World War II. Credito Emiliano's two climate-controlled warehouses hold about 440,000 wheels worth €132 million, or $187.5 million.
>
> "This mechanism is our life blood," said Giuseppe Montanari, a cheese producer and dealer who uses the loans to buy milk. "It's a great way to finance our expenses at convenient rates, and the bank doesn't risk much because they can always sell the cheese."

### The Gold Standard

After thousands of years, two commodities beat all of the others in terms of maintaining their monetary attributes across multiple geographies; gold and silver. Only they were able to retain a high enough stock-to-flow ratio to serve as money, despite civilizations constantly improving their technological capabilities throughout the world over the ages.

Humans figured out how to make or acquire basically all of the beads, shells, stones, feathers, salt, furs, livestock, and industrial metals we need with our improved tools, and so we reduced their stock-to-flow ratios and they all fell out of use as money.

However, despite all of our technological progress, we still can't reduce the stock-to-flow ratios of gold and silver by any meaningful degree, except for rare instances in which the developed world found new continents to draw from. Gold has maintained a stock-to-flow ratio averaging between 50 and 100 throughout modern history, meaning we can't increase the existing supply by more than about 2% per year, even when the price goes up more than 10x in a decade. Silver generally has a stock-to-flow ratio of 10 to 20 or more.

Most other commodities are below 1 for the stock-to-flow ratio, or are very flexible. Even the other rare elements, like platinum and rhodium, have very low stock-to-flow ratios due to how rapidly they are consumed by industry.

We've gotten better at mining gold with new technologies, but it's inherently rare and we've already tapped into the "easy" surface deposits. Only the deep and hard-to-reach deposits remain, which acts like an ongoing difficulty adjustment against our technological progress. One day we could eventually break this cycle with drone-based asteroid mining or ocean floor mining or something crazy like that, but until that day (if it ever comes), gold retains its high stock-to-flow ratio. Those environments are so inhospitable that the expense to acquire gold there would likely be extraordinarily high.

Basically, whenever any commodity money came into contact with gold and silver as money, it was always gold and silver that won. Between those two finalists, gold eventually beat silver for more monetary use-cases, particularly in the 19th century.

Improvements in communication and custody services eventually led to the abstraction of gold. People could deposit their gold into banks and receive paper credit representing redeemable claims on that gold. Banks, knowing that not everyone would redeem their gold at once, went ahead and issued more claims than the gold they held, beginning the practice of fractional reserve banking. The banking system then consolidated into central banking over time in various countries, with nationwide slips of paper representing a claim to a certain amount of gold.

Barry Eichengreen's explanation for why gold beat silver, in his book Globalizing Capital: A History of the International Monetary System, is that the gold standard won out over the bimetallic standard mostly by accident. In 1717, England's Master of the Mint (who was none other than Sir Isaac Newton himself) set the official ratio of gold and silver as it relates to money, and according to Eichengreen he set silver too low compared to gold. As a result, most silver coins went out of circulation (as they were hoarded rather than spent, as per Gresham's law).

Then, with the UK rising to dominance as the strongest empire of the era, the network effect of the gold standard, rather than the silver standard, spread around the world, with the vast majority of countries putting their currencies in a gold standard. Countries that stuck to the silver standard for too long, like India and China, saw their currency weaken as demand for the metal dropped in North America and Europe, resulting in negative economic consequences.

On the other hand, Saifadean Ammous, in his book The Bitcoin Standard, focuses on the improved divisibility of gold due to banking technology. As previously mentioned, gold scores equal or higher than silver in most of the attributes of money, except for divisibility. Silver is better than gold for divisibility, which made silver the more "day to day" money for thousands of years while gold was best left for kings and merchants to keep in their vaults or use as ornamentation, which are stores and displays of value respectively.

However, the technology of paper banknotes in various denominations backed by gold improved gold's divisibility. And then, in addition to exchanging paper, we could eventually "send" money over telecommunications lines to other parts of the world, using banks and their ledgers as custodial intermediaries. This was the gold standard- the backing of paper currencies and financial communication systems with gold. There was less reason to use silver at that point, with gold being the much scarcer metal, and now basically just as divisible and even more portable thanks to the paper/telco abstraction.

I think there is an element of truth in both explanations, although I consider the explanation of Ammous to be more complete, starting with a deeper axiom regarding the nature of money itself. Banknotes made gold more divisible and thus the harder money won out over time, but network effects from political decisions can impact the timing of these sorts of changes.

Central banks around the world still hold gold in their vaults, and many of them still buy more gold each year to this day as part of their foreign-exchange reserves. It's classified as a tier one asset in the global banking system, under modern banking regulations. Thus, although government-issued currency is no longer backed by a certain amount of gold, it remains an indirect and important piece of the global monetary system as a reserve asset. There is so far no better naturally-occurring commodity to replace it.

Gold used to trade at a 10x to 20x multiple of silver's value for thousands of years in multiple different geographies. Over the past century, however, the gold-to-silver price ratio has averaged over 50x. Silver seems to have structurally lost a lot of its historical monetary premium relative to gold over the past century. Chart via Longtermtrends.net:




If you prefer listening to reading for this section, I had a February 2022 discussion with Stig Brodersen about the history of gold and commodities.

# Fiat Currency

Historically, a number of cultures have attempted periods of paper currency, issued by the government and backed by nothing.

Often it was the result of currency that was once backed (a gold standard or silver standard), but the government created too much of the paper due to war or other issues, and had to default on the metal backing by eliminating its ability to be converted back into the metal upon request. In that sense, currency devaluation becomes a form of tax and/or wealth confiscation. The public holds their savings in the paper currency, and then the rug is pulled out

The general argument for why fiat currencies exist, is that most governments, if possible, do not want to be constrained by gold or other scarce monies, and instead want to have more flexibility with their spending.

The earliest identified use of paper currency was in China over a thousand years ago, which makes sense considering that paper was invented in that region. They eventually shifted towards government monopoly on paper currency, and combined with an elimination of its ability to be converted back into silver, resulted in the first fiat currency, along with the inflation that comes with that. It didn't last very long.

Fiat currency is interesting, because unlike the history of commodity money, it's a *step down* in terms of scarcity. Gold beat out all of the other commodity monies over centuries of globalization and technological development, and then gold itself was defeated by… pieces of paper?

This is generally attributed to technology and government power. As clans became kingdoms, and as kingdoms became nation states, along with the creation of banking systems and improvements in communication systems, governments could become a larger part of everyday life. Once gold became sufficiently centralized in the vaults of banks and central banks, and paper claims were issued against it, the only remaining step was to end the redeemability of that paper and enforce its continued usage through legal obligation.

**Debasing Currency and Empowering Wars**

Currency debasement often happened gradually under metallic and bimetallic currency regimes, with history of it going back three or four thousand years. It took the form of reducing the amount of the valuable metal (such as gold or silver) and either adding base metal or putting decorative holes through the center of it to reduce the weight.

In other words, a ruler often found himself faced with budget deficits, and having to make the difficult choice between cutting spending or raising taxes. Finding both to be politically challenging, he would sometimes resort to keeping taxes the same, diluting the content of gold or silver in the coins, and spending more coins with less precious metal in each coin, while expecting it to still be treated with the same purchasing power per coin.

For example, a king can collect a 1,000 gold coins in taxes, melt them down and make new coins that are each 90% gold (with the other 10% made from some cheap filler metal), and spend 1,111 gold coins back into the economy with the same amount of gold. They do look pretty similar to most people, but some discerning people will notice. Years later, if that's not enough, he could re-melt them and make them 80% gold, and spend 1,250 of them into the economy…

At first, these slightly-debased coins would be treated as how they were before, but as the coins are increasingly debased, it would become obvious. Peoples' savings would decline in value, as they found over time that their stash of gold and silver was only fractional gold and silver. Foreign merchants in particular would be quick to demand more of these debased gold coins in exchange for their goods and services.

Gold-backed paper currencies and fiat currencies are the modern version of that, and so the debasement can happen much faster.

At first, fiat currencies were created temporarily, in times of war. After the shift from commodity money to gold-backed paper, the gold-backing would be briefly suspended as an emergency action for a number of years, and then re-instated (usually with a significant devaluation, to a lower amount of gold per unit of currency, since a lot of currency was issued during the emergency period).

This is a faster and more efficient way to devalue a currency than to actually debase the metal. The government doesn't have to collect everyone's coins and re-melt them. Instead, everybody is holding paper money that they trust to be redeemable for a certain amount of gold, and the government can break that trust, suspend that redeemability, print a ton of paper money, and then re-peg that paper money such that each unit of paper currency is redeemable for a much smaller amount of gold, before people realize what is happening to their savings.

That method instantly debases peoples' money while they continue to hold it, and can be done overnight with the stroke of a pen.

Throughout the 20th century, this tactic spread around the world like a virus. Prior to paper currencies, governments would run out of fighting capability if they ran low on gold. Governments would use up their gold reserves and raise additional war taxes, but there were limits in terms of how much gold they had and how much they could realistically tax for unpopular wars before the population would rebel. However, by having all of their citizens on a gold-backed paper currency, they could devalue everyone's savings for the war without an official tax, by printing a lot of money, spending it into the economy, and then eliminating or reducing the gold peg before people knew what was happening to their money.

This allowed governments to fight much larger wars by extracting more savings from their citizens, which led their international opponents to debase their currencies with similar tactics as well if they wanted to win.

Ironically, the fact that fiat currencies have no cost to produce, is what gave them the biggest cost of all.

### Bretton Woods and the Petrodollar

After World War I, and throughout the tariff wars and World War II period thereafter, many countries went off the gold standard or devalued their currencies relative to gold.

John Maynard Keynes, the famous economist, said in 1924:

> “ In truth, the gold standard is already a barbarous relic.

By 1934, gold was made illegal to own. It was punishable by up to 10 years in prison for Americans to own it. The dollar was no longer redeemable for gold by American citizens, although it was still redeemable for official foreign creditors, which was an important part of maintaining the dollar’s credibility.

Shortly after Americans were forced to sell their gold to the government in

the government at the expense of those who were forced to sell it.

It remained illegal for Americans to own gold for about four decades until the mid-1970s. Interestingly enough, that overlapped quite cleanly with the period where US Treasuries underperformed inflation. Basically, the main release valve that people could turn to instead of cash or Treasuries as savings assets, was made illegal to them:

It's rather ironic- gold was a "barbarous relic" and yet apparently had to be confiscated and pushed out of use by the threat of imprisonment, and hoarded only by the government during a period of intentional currency devaluation. If it were truly such a relic, it would have fallen out of usage on its own and the government would have had little need to own any.

looking for it.

By 1944 towards the end of World War II after most currencies were sharply devalued, the Bretton Woods agreement was reached. Most countries pegged their currency to the dollar, and the United States dollar remained pegged to gold (but only redeemable to large foreign creditors, not American citizens). By extension, a pseudo gold standard was temporarily re-established.

This lasted only 27 years until 1971, when the United States no longer had enough gold to maintain redemption for its dollars, and thus ended the gold standard for itself and most of the world. There were too many dollar claims compared to how much gold the US had:

The Bretton Woods system was poorly-constructed from the beginning, because domestic and foreign banks could lend dollars into existence without having to maintain a certain amount of gold to back those dollars. The mechanism for dollar creation and gold were completely decoupled from each other, in other words, and so it was inevitable that the quantity of dollars in existence would quickly outpace how much gold the US Treasury had in its vaults. As the amount of dollars multiplied and the amount of available gold did not, any smart foreign creditor would begin redeeming dollars for gold and draining the Treasury's vaults. The Treasury would be quickly drained of its gold until they either sharply devalued the dollar peg or ended the peg altogether, which they did.

Since that time, over 50 years now, virtually all countries in the world have been on a fiat currency system, which is the first time in history this has happened. Switzerland was an exception that kept their gold standard until 1999, but for most countries it has been over 50 years since they were on it.

However, the US dollar still has a vestige of commodity-backing, which is part of what kept this system together for so long. In the 1970s, the US made a deal with Saudi Arabia and other OPEC countries to only sell their oil in dollars, regardless of which country was buying. In return, the US would provide military protection and trade deals. And thus the petrodollar system was born. We've had to deal with the consequences of this awkward relationship ever since.

While the dollar was not pegged to any specific price of oil in this system, this petrodollar system made it so that any country in the world that needed to import oil, needed dollars to do so. Thus, universal demand for dollars was established, as long as the US had enough military might and influence in the Middle East to maintain the agreement with the oil exporting nations.

Other countries continued to issue their own currencies but held gold, dollars

reserves to back up their currencies. Most of their currencies were not pegged to any specific dollar, oil, or gold value during this time, but having a large reserve that they could use to actively maintain the strength of their currency was a key part of why global creditors would accept their currency.

The biggest benefit from the petrodollar system, as analyst Luke Gromen has argued, is that it contributed to the US's Cold War victory over the Soviet Union during the 1970s and 1980s. The petrodollar agreement and associated military buildup to enforce it was a strong chess move by the US to gain influence over the Middle East and its resources. However, Gromen also argues that when the Soviet Union fell in the early 1990s, the US should have pivoted and given up this system to avoid ongoing structural trade deficits, but did not, and so its industrial base was aggressively hollowed out. Since then, China and other countries have used the system *against* the US, and the US also bled out tremendous resources trying to maintain its hegemony in the Middle East with its wars in Afghanistan and Iraq.

An international gold standard looks like this, with each major country pegging its own currency to a fixed amount of gold and holding gold in reserve, for which it was redeemable to its citizens and foreign creditors:

The Bretton Woods pseudo gold standard involved the dollar being backed by gold, but only redeemable to foreign creditors in limited amounts. Foreign currencies pegged themselves to the dollar, and held dollars/Treasuries and gold in reserve:

The petrodollar system made it so that only dollars could buy oil imports around the world, and so countries globally hold a combination of dollars, gold, and other major currencies as reserves, with an emphasis on dollars. If countries want to strengthen their currencies, they can sell some reserves and buy back their own currency. If countries want to weaken their currencies, they can print more of their currency and buy more reserve assets.

Over time, that demand for dollars was broadened via trade and debt. If two countries trade goods or services, they often do so in dollars. When loans are made internationally, they are often done so in dollars, and the world now has over $13 trillion in dollar-denominated debt, owed to all sorts of places including lenders in Europe and China. All of that dollar-denominated debt represents additional *demand* for dollars, since dollars are required to service that debt. Basically, the petrodollar deal helped initiate and maintain the network effect at a critical time, until it became rather self-sustaining.

This system gives the United States considerable geopolitical influence, because it can sanction any country and cut it off from the dollar-based system.

One of the key flaws of the petrodollar system, however, is that all of this demand for the dollar makes US exports more expensive (less competitive) and makes imports less expensive, and so the US began running structural trade deficits once we established the system, totaling over $14 trillion in cumulative deficits as of this writing. From 1944-1971 the US drew down its gold reserves in order to maintain the Bretton Woods dollar system, whereas from 1974-present, the US instead drew down its industrial base to maintain

*Chart Source: Trading Economics*

As the FT described in a clever article back in 2019, this petrodollar system ironically gave the United States a form of Dutch Disease. For those who aren't familiar with the term, Investopedia has a good article on Dutch Disease. Here's a summary:

> The term Dutch disease was coined by The Economist magazine in 1977 when the publication analyzed a crisis that occurred in the Netherlands after the discovery of vast natural gas deposits in the North Sea in 1959. The newfound wealth and massive exports of oil caused the value of the Dutch guilder to rise sharply, making Dutch exports of all non-oil products less competitive on the world market. Unemployment rose from 1.1% to 5.1%, and capital investment in the country dropped.
>
> Dutch disease became widely used in economic circles as a shorthand way of describing the paradoxical situation in which seemingly good news, such as the discovery of large oil reserves,

As the FT argues (correctly in my view), making virtually all global oil priced in dollars basically gave the United States a form of Dutch Disease. Except instead of finding oil or gas, we engineered a system so that every country needs dollars, and so we need to export a lot of dollars via a structural trade deficit (and thus, the dollar as a global reserve asset basically served the role of a big oil/gas discovery).

This system, much like the Netherlands' natural gas discovery, kept US currency persistently stronger at any given time than it should be on a trade balance basis. This made actual US exports rather uncompetitive, boosted our import power (especially for the upper classes) and prevented the US balance of trade from ever normalizing for decades.

Japan and Germany became major exporters at our expense, and for example, their auto industries thrived globally while the US auto industry faltered and led to the creation of the "Rust Belt" across the midwestern and northeast part of the country. And then China grew and did the same thing to the United States over the past twenty years; they ate our manufacturing lunch. Meanwhile, Taiwan and South Korea became the hubs of the global semiconductor market, rather than the United States.

That petrodollar system is starting to crack under its own weight, as trade deficits have collected into a massively negative net international investment position for the US, and the US has more wealth concentration than the rest of the developed world because we hollowed out a lot of our blue collar workforce specifically. This causes rising political tensions and desires (so far unsuccessful) to reduce the trade deficit and rebuild our industrial base. Foreigners take their persistent dollar surpluses and buy productive US assets with them like stocks, real estate, and land. In other words, the US sells its appreciating financial assets in exchange for depreciating consumer goods:

My article on the petrodollar system went into additional detail on the history of the US dollar as the global reserve currency, from the pre-Bretton Woods era to the petrodollar system.

**Potential Post-Petrodollar Designs**

There are proposals by policymakers and analysts to rebalance the global payments system, and the changing nature of geopolitics is pointing in that direction as well.

For example, Russia began pricing its oil partly in euros over the past few years, and China has put considerable work into launching a digital currency that may expand their global reach, at least with some of their most dependent trade partners. The United States is no longer the biggest commodity importer, and its share of global GDP continues to decrease, which makes the existing petrodollar system less tenable.

If several large fiat currencies can be used to buy oil, then the model looks more like this (and many tertiary currencies would manage themselves relative to these main currencies that have the scale and influence to buy oil

If a major scarce neutral reserve asset (e.g. gold or bitcoin or digital SDRs or something along these lines, depending on your conception of where trends are going over the next decade or two) is used as a globally-recognized form of money, then a decentralized model can also look like this:

Overall, it's clear that the trend in global payments is towards digitization and decentralization away from one single country's currency, but it's unclear precisely how the next system will turn out and on what timeline it will

## Price Inflation from a Negative Baseline

The long arc of human history is deflationary. As our technology improves over time, we become more productive, which reduces the labor/resource cost of most goods and services. This is particularly true over the past couple centuries as humanity exponentially tapped into dense forms of energy. Prior to that, our rate of productivity growth was much slower.

For an example of productivity, people used to farm by hand. By harnessing the utility of work horses and simple equipment, it empowered one person to do the work of several people. Then, the invention of the tractor and similar advanced equipment empowered one person to do the work of ten or more people. As tractor technology got bigger and better, this figure probably jumped to thirty or more people. And then, we can imagine a fleet of self-driving farming equipment allowing one person to do the work of a hundred people. As a result, a smaller and smaller percentage of the population needs to work in agriculture in order to feed the whole population. This makes food less expensive and frees up everyone else for other productive pursuits.

Gold has historically appreciated against most other commodities over time, like an upward sine wave. Alternatively, we can say that most commodities depreciated in price against gold like a downward sine wave. For example, there are inflationary cycles where copper increases in price compared to gold, but over multiple decades of cycles, gold has steadily appreciated against copper. For agricultural commodities that are less scarce, the trend is even stronger.

Here's a chart of the copper-to-gold ratio showing its structural decline and cyclical exceptions since 1850, for example:

*Chart Source: Long Term Trends*

And here's wheat priced in gold since 1910:

*Chart Source: Priced in Gold*

This is because over time, our advancing technology has made us more efficient at harvesting those other commodities. However, gold's extreme scarcity and rather strict stock-to-flow ratio of over 50x has made it so that our technology advancements in finding and mining gold are offset by the fact that we've already mined the "easy" gold deposits and the remaining deposits are getting deeper and harder. We never truly get more efficient at retrieving gold, in that sense. It's a built-in ongoing difficulty adjustment.

During the late 1800s and early 1900s USA, which is when the country became a rising power globally, the country was on a gold standard and in a period of structural deflation. Prices of most things went down because land was abundant and huge advancements in technology during the industrial age made people far more productive.

televisions exponentially better and cheaper over time, especially when priced in gold. Similarly, cell phones decades ago were very large, basic, and expensive toys for the wealthy. Now, many people in the poorest regions of the world have powerful smartphones as a normal course of life. They have supercomputers in their pockets.

Overall, we can say that the baseline inflation rate is some negative number (aka deflation), and how negative it is at any given time depends on the pace of technological advancement. Baseline inflation only becomes a positive number if we are backtracking in some way, and thus encountering more scarcity and less abundance. This could be due to malinvestment or war, for example.

By holding the most salable good (such as gold, historically), your purchasing power gradually appreciates over time because the labor/resource cost of most other things goes down whereas that salable good retains most or all of its scarcity and value. The vast majority of commodities, products, and services structurally decrease in price gradually relative to your strong store of value.

One way to measure this is by looking at the broad money supply per capita over time relative to the consumer price index. Here's a chart of their 5-year rolling average growth rates for the United Kingdom:

We can see that there is usually a positive gap between broad money supply growth per capita, and consumer prices. Broad money supply per capita increased by an average of 5.3% per year during this 150+ year period, while a basket of goods and services increased by an average of only 3.1%. In other words, monetary inflation is usually a bit faster than price inflation.

In a very rough sense by this way of looking at it, real productivity growth was about 2.2% per year, which is the difference between these figures. What this means is that in any given year, the resource/labor costs of a broad basket of goods and services goes down by an average of 2.2% due to technological progress, but the amount of money that people have goes up by 5.3%, and as

So, price inflation is not 3.1% from a baseline of zero; it's 5.3% from a baseline of -2.2%. Actual resource costs for goods and services go down most years rather than stay flat, but due to our inflationary monetary framework, they go up in price anyway.

The reason this is only a rough measure is because 1) the CPI basket changes over time and may not be fully representative and 2) money supply can become more concentrated or less concentrated over time and thus does not always reflect the buying power of the median person. There is no way to directly measure technological deflation; it can only be estimated.

Another way to check this is to simply see to what extent gold appreciated in price against the British pound, and the answer is about 4.0% per year over this same 150+ year time period. Gold appreciated in price faster than the CPI basket inflation rate by about 0.9% per year (the difference between 4.0% and 3.1%, which compounds quite a bit over a century), and can buy you a bit more copper, oil, wheat, or many other goods and services than it could 50, 100, or 150 years ago, unlike the British pound which buys you a lot less than it used to. Higher-quality and scarcer goods like meat have roughly kept up with the price of gold (although you can't store meat for very long), and select few assets like the absolute best/scarcest UK property locations may have appreciated a bit faster than gold (although they required continued maintenance costs along the way which makes up for that difference).

The takeaway from this section is that the growth in the broad money supply per capita is the "true" inflation rate. However, the baseline that we measure it against is not zero; it's a mildly negative number which we can't precisely measure, but that we can estimate and infer, that represents ongoing increases in productivity due to technology. Prices of most things stay relatively stable or preferably keep going *down* as priced in the most salable good (such as gold, historically) over the long run, but go up in most years when measured in a depreciating and weaker unit of account such as the

## The MMT Description of Fiat Currency

Some economists disagree with the commodity view of money, and argue that money originates with the government. This is called Chartalism, and its origins go back more than a century.

Decades ago, Warren Mosler and others resurfaced this idea, into what is now popularly known as Modern Monetary Theory or MMT.

I have often felt that Mosler describes the case for that school of thought well. He doesn't sugar-coat things, and instead speaks very directly:

> Start with the government trying to provision itself and how does it do it. There are different ways to do it throughout history. One way is just to go out and take slaves. Another way the British did is they provisioned their navy by going to bars late at night and dragging them onto ships. It's called impressing sailors.
>
> We pretend to be more civilized as I like to say, and we use a monetary system. So how does a government do it? Clean sheet of paper, what you do, is you establish a tax that is payable in something that people don't have. So what you want to do is transfer resources from the private sector to the public sector. You want people who are out there doing whatever they are doing, to suddenly be working for the government. You need soldiers, you need police, you need health workers, you need people in education. How do you get these people out of the private sector and into the public sector?
>
> First thing you do is you levy a tax. You need a tax liability, and it has to be coercive. And for this example I'll use a property tax. You put a tax on everybody's house, and you make it payable in your

thing that is used to pay the tax. The dollar, the yen, or the euro- they are all tax credits.

What has happened is, you've created sellers of real goods and services who now need your tax credit, or they're going to lose their house. You've created unemployment- people looking for paid work. Unemployment is not about people looking to volunteer at the American Cancer Society; it's about people looking for work because they need or want the money. And the problem with government when you want to provision yourself is that there's no unemployment. No unemployment in terms of your currency; there may be people willing to work for other things but not for your currency. You need unemployment in terms of your currency, looking to earn your unit of account.

So you levy a tax, now people need your unit of account, all of these people show up looking for work, all of those people are unemployed. You now hire the unemployed that your tax created, and they are now provisioning your government.

-Warren Mosler, 2017 MMT Conference

I also liked this description, where he explained his view in an economic debate:

> The way we do it, is we slap on a tax for something that nobody has, and in order to get the funds to pay that tax, you have to come to the government for it, and so that way the government can spend its otherwise worthless currency and provision itself.
>
> Now the way I like to explain that, is I'll take out my business card

> one of these cards for a hundred dollars? No? Okay. Does anybody want to stay after hours and help vacuum the floor and clean the room and I'll give you my cards? No? Alright. Oh by the way there's only one door out of here and my guy is out there with a 9mm (handgun) and you can't get out of here without one of these cards.
>
> Can you feel the pressure now? You're now unemployed! In terms of my cards, you were not unemployed before. You were not looking for a job that paid in my cards. Now you're looking for a job that pays in my cards, or you're looking to buy them from someone else that will take a job that pays in my cards.
>
> [...]
>
> The difference between money and litter is whether there's a tax man [outside that door]. The guy with the 9mm is the tax man. If he can't enforce tax collection, the value of the dollar goes to zero.
>
> -Warren Mosler, 2013 MMT vs Austrian Debate

Nobel-laureate economist Paul Krugman put it rather similarly back in 2013:

> " Fiat money, if you like, is backed by men with guns.

Of course, we could just as easily ask, since the government is using force to collect taxes to provision itself, why can't it just collect commodity money like gold with a tax, and then spend that gold to acquire its necessary provisions? Why does it need to issue its own paper currency and then tax it back?

The answer is that it doesn't have to, but it wants to. By issuing its own currency, it profits from seigniorage, which is the difference between the face

value of the money and the cost to produce and distribute it. It is, basically, a subtle inflation tax that compounds over time.

A weak government, with an economy that can't provision most of its needs, often fails to maintain a workable fiat currency for very long. People start using alternative monies out of necessity even if the government supposedly disallows them from doing so. This happens to many developing countries. Billions of people in the world today have experienced the effects of hyperinflation or near-hyperinflation within the past generation. It's unfortunately quite common.

However, developed countries have been more successful at maintaining seigniorage over the past five decades of the fiat system. Their currencies all lost 95% to 99% of their purchasing power over time, but it was usually gradual rather than abrupt. The system is not without its cracks as previously-discussed, but it's certainly the most comprehensive fiat currency system ever constructed.

When optimized skillfully, a fiat currency has low volatility year-to-year, in exchange for gradually losing value over the long run. By being actively managed with taxes, spending, and central bank reserve management (creating or destroying currency in exchange for reserve assets), policymakers try to maintain a low and steady inflation rate, meaning a mild and persistent decline in the purchasing power of their currency.

A strong government can force the use of its currency over all other types of monies within its borders, at least for a medium of exchange (not necessarily a store of value), by taxing other types of transactions, and by only accepting its fiat currency as a form of payment for taxes. They can make things like gold, silver, and bitcoin less convenient as money, for example, by making each transaction with it a taxable event in terms of capital gains. If push comes to shove, they can also try to ban those things with threat of force.

### The Monetization of Other Assets

Although most of us today are used to it, fiat currency has been a polarizing and inherently political subject ever since the world went onto this petrodollar standard five decades ago.

Mainstream media and economists, however, quickly adopted it as canon and rather unquestionable. For decades now, it has been the case that if someone thinks money shouldn't be fiat currency, they're kind of considered a kook and not taken seriously. This kind of vibe:

But when you step back and think about it from first principles, this period in history is really unusual. It's a historical aberration, and like a fish in water doesn't even notice the water, the monetary system we operate with now seems totally normal to us.

Never before, in thousands of years of human history, has the entire world been using a money that has no resource cost or constraint. It's an experiment, in other words, and we're five decades into it. Many would consider it a good experiment, while others consider it a bad one, but it's not as though it is inevitable, or the only possible outcome from here. It's simply what we have now, and who knows what things will look like in another five decades.

To put it into perspective, this international monetary system based around centrally-managed fiat currency is only 16 years older than me. My father was 36 when the US went off the gold standard. When I grew up, after a period of financial hardship, I began collecting gold and silver coins as a kid; my father gave me silver coins as savings each year.

The Swiss dropped their gold standard when I was twelve years old, which was six years after Amazon was founded, and three years before Tesla was founded. The fiat/petrodollar standard is only four times older than bitcoin, and only two times older than the first internet browser. That's pretty recent when you think about it like that.

Ever since the world has been on the fiat/petrodollar standard, debt as a percentage of GDP has skyrocketed to record levels and seems to be getting unstable. Considering where we are in the long-term debt cycle, investors would do well to be creative with how they envision the future. Don't take the past 40-50 years for granted and assume that's how it'll always be, whether for money or anything else. We don't know what money will look like 50 years from now.

The last time we were in a similar debt and monetary policy situation was the 1930s and 1940s, where currency devaluations and war occurred. That doesn't mean those things have to happen, but basically, we're in a very macro-heavy environment where structural currency changes tend to occur.

One of the results of fiat currency, especially towards the later stages of this five decade experiment since the 1970s, is that more people have begun to treat cash like a hot potato. We instinctively monetize other things, like art, stocks, home equity, or gold. The ratio of home prices to median income has gone up a lot, as well as the ratio of the S&P 500 to median income, or a top-notch piece of art to median income.

This chart shows the loss of purchasing power of the U.S. dollar since the Coinage Act of 1792, which is when the US dollar and the US Mint were created:

*Chart Source: Ian Webster, annotated by Lyn Alden*

It currently takes nearly $3,000 to have as much purchasing power as $100 bought in 1792. From 1792 to 1913, the dollar's purchasing power oscillated mildly around the same value, with over 120 years of relative stability. From

And it's actually worse today than during most of this 1971-2022 fiat/petrodollar period, because interest rates aren't keeping up with inflation rates anymore. The fiat system is getting less stable due to so much debt being in the system, which disallows policymakers from raising interest rates higher than the prevailing inflation rate.

Basically, for lack of good money in this fiat currency petrodollar era, especially in the post-2009 era with interest rates below inflation rates, we monetize other things with higher stock-to-flow ratios and treat them as stores of value.

In China, consumers aggressively monetize real estate. It became normal for families to own multiple homes. In the United States, consumers aggressively monetize stocks. We plow a percentage of each paycheck into broad equity indices without analyzing companies or doing any sort of due diligence, treating that basket of stocks as simply a better store of value than cash regardless of what is inside.

We can ask, for example, would we rather own dollars that went from 10 trillion in quantity ten years ago to 22 trillion today and pay basically no yield to own them, or Apple shares that went from 26 million shares ten years ago to 16 million shares today and also pay basically no yield to own it? Is the dollar better money, or is a diverse collection of fungible corporate shares better money, when it comes to storing value with a 5+ year time horizon?

This monetization of non-money securities and property opens us up to more volatility, more leverage, less liquidity, less fungibility, and more taxable events. Basically, rather than investments being special things we make with careful consideration, we shovel most of our free capital into hundreds of index investments that we don't even analyze, since who would hold currency for any significant length of time? Fungible pieces of corporations become our money, at least for the "store of value" portion of what money is, in large part because they pay higher earnings/dividend yields than bank/bond yields and many of them decrease in quantity (deflationary) rather than constantly increase in quantity.

Some technologists, like Jeff Booth, have argued that this system of perpetual currency debasement has a negative impact on the environment because it encourages us to spend and consume on short-sighted depreciating trinkets and malinvestment more than we would if our money appreciated in value over time like it used to. With appreciating money, we would be more selective with our purchases.

Proponents of the fiat system argue that it smooths out economic downturns and allows for counter-cyclical investment and stimulus. By having a flexible monetary base, policymakers can increase or decrease the supply of money in order to provide a balancing force against credit cycles and industrial output capacity. In exchange for a persistently declining value of currency, we get a more stable currency on a year-to-year basis.

In addition, proponents of the system also argue that the system encourages more consumption and consider that to be a good thing because it keeps GDP up. By keeping people on a constant treadmill of currency debasement, it forces them to spend and invest rather than to save. If people begin to save, these policymakers often view it as "hoarding" or a "global savings glut" and consider it to be a problem. Monetary policy then is adjusted to convince people to save less, spend more, and borrow more.

From a developing market standpoint, the fiat/petrodollar standard contributes to massive booms and busts because a lot of their debt is denominated in dollars, and that debt fluctuates wildly in strength depending on the actions of US policymakers. Developing countries are often forced to tighten their monetary policy during a recession in order to defend their currency, and thus while the US gets to provide counter-cyclical support for its own economy, developing countries are forced to be pro-cyclical, contributing to a vicious cycle in their economies during recessions. In this view, the fiat/petrodollar system can be considered a form of neocolonialism; we push most of the costs of the system out into the developing countries in

Overall, the fiat system is showing more instability lately, and investors have to navigate a challenging environment of structurally negative inflation-adjusted cash and bond yields, along with many high asset valuations in equities and real estate.

### Sovereign International Reserves

As countries accumulate trade surpluses, they keep those gains in sovereign international reserves. This represents the pool of assets that a country's central bank can draw upon to defend the country's currency if needed. The more reserves a country has relative to its GDP and money supply, the more defense it has against a meltdown in its fiat currency. The country can sell these reserves and buy back its own currency to support its currency per-unit value. The currency may not be backed up by gold at a redeemable rate, but it's backed up by diverse assets as needed if its starts to rapidly lose value.

The world collectively has about $15 trillion USD-equivalents worth of official sovereign reserves. Less than $2.5 trillion of that is gold, with more than $12.5 trillion held as fiat reserves (dollars, euros, yen, francs, etc). Fiat reserves consist of government bonds and bank deposits, and can be easily frozen by the countries that issue them. In addition, many gold holdings are not held within the country, but instead are held in New York or London on their behalf.

Thus, the vast majority of sovereign official reserves, are permissioned assets rather than permissionless assets. They are non-sovereign; able to be frozen by foreign nations. War crystalizes this fact.

In February 2022, Russia invaded Ukraine. Russia had $630 billion USD-equivalents of sovereign international reserves prior to the war, representing decades of accumulated trade surpluses as sovereign savings to underpin their currency. Of this $630 billion, $130 billion consisted of gold, and the

maybe $70-$80 billion consisted of Chinese fiat assets, and the other $400+ billion consisted of European and other fiat assets. Europe subsequently froze that $400+ billion in Russian fiat assets in response to Russia's invasion of Ukraine, which is equivalent to over 20% of Russian GDP and over five years of Russian military spending; an utterly massive economic confiscation. Russia is currently in a financial crisis, and it remains to be seen if they can exert enough commodity/military pressure to have their reserves unfrozen.

Some could argue that it's a good thing that countries hold their reserves in each other's assets and thus can be frozen. Along with trade sanctions, this practice gives countries another lever with which to control each others' behavior away from extremes (such as war). We're all interdependent to one degree or another anyway. But from a pragmatic point of view, countries tend to want to reduce their vulnerabilities and external risks where possible, and that can include minimizing the ability of their stored-up central bank reserves to be confiscated or frozen by other countries.

I began writing this longform article months ago, in late 2021. Things have accelerated since then, and for example the WSJ ran an article in early March 2022 called "If Russian Currency Reserves Aren't Really Money, the World is in for a Shock." Here's the opening paragraph:

> "What is money?" is a question that economists have pondered for centuries, but the blocking of Russia's central-bank reserves has revived its relevance for the world's biggest nations—particularly China. In a world in which accumulating foreign assets is seen as risky, military and economic blocs are set to drift farther apart.

What is money?

Well, the answer to that question ties into the difference between currency

whether or not to honor that particular liability. Money is something that is intrinsically valuable in its own right to other entities, and that has no counterparty risk if you custody it yourself (although it may have pricing risk related to supply and demand). In other words, Russia's gold is money; their FX reserves are currency. The same is true for other countries.

Fiat currency and government bonds have no intrinsic value; they represent indirect claims of value that can be blocked and confiscated. Gold has value; it's sufficiently fungible and due to its physical properties, various entities would accept gold at the current market value. It can be self-custodied and no external nation can shut it off.

Currency acts like money most of the time until, one day, it doesn't.

**Fiat Summarized**

Overall, the key feature or bug of fiat currency (depending on how you look at it) is its flexible supply and its ability to be diluted. It allows governments to spend more than they tax, by diluting peoples' existing holdings. With this feature, it can be used to re-liquify seized-up financial situations, and stimulate an economy in a counter-cyclical way. In addition, its volatility can be minimized compared to commodity monies most of the time through active management, in exchange for ensuring gradual devaluation over time.

When things go wrong, however, fiat currency can lose value explosively. Fiat currency tends to incentivize running bigger deficits (since spending doesn't necessarily need to be taxed for), and generally requires some degree of hard or soft coercion in order to get people to use it over harder monies, although that coercion is often rather invisible to most people most of the time, until things go wrong. And its ability to be diluted can allow for longer wars, selective bailouts for influential groups, and other forms of government spending that aren't always transparent to citizens.

# Digital Assets

With the development of the internet and cryptography in the 1980s and 1990s, many people began working on internet-native money systems. Hash Cash, Bit Gold, and B-Money were some early examples.

Some of these early pioneers wanted to be able to easily pay on the internet, which wasn't quite as easy back then. Others were part of the cypherpunk movement: people who responded to the information age and the lack of privacy it would increasingly bring, by advocating for transactional privacy through encryption.

Freedom House, a nonprofit organization founded in 1941 and originally chaired by Eleanor Roosevelt, has indeed noted that authoritarianism has been on the rise in recent decades. More than half the world's population lives in an authoritarian or semi-authoritarian country. People in privileged areas often fail to recognize this trend.

*Chart Source: Freedom House*

The world became more free in the 1980s and 1990s as China and the Soviet Union opened up, but then the world increasingly began chipping away at that freedom in the 2000s and 2010s, at least as far as Freedom House and various other sources measure it. China in particular is a huge surveillance and control state now, with transactions and online behavior monitored and organized, social credit scores determined from the data, and consequently near-complete control over their citizens' behavior.

Even the developed world began introducing policies that chipped away at

countries mildly declined over time as well. For example, the United States was ranked 94 for its freedom score back in 2010, but as of 2020 was ranked only 83. It has been reported for over a decade now, with increasing revelations over time, that the CIA and NSA have large spying operations on Americans.

The more digital the world is, the more authoritarian regimes, semi-authoritarian regimes, or would-be-authoritarian regimes are able to monitor and intervene in their subjects' lives. Authoritarianism combined with 21st century digital surveillance technology and Big Data to organize it all, is a rather frightening prospect for a lot of people. This combination has been predicted by science fiction books for decades.

**The Discovery of Digital Scarcity and the Invention of Bitcoin**

The ability to transact with others is a key part of individual liberty. The more that authoritarian regimes can control that, the more power they have over their citizens' lives.

Nobel-laureate economist Friedrich Hayek once gave an interesting statement on the subject of money:

> "I don't believe we shall ever have a good money again before we take the thing out of the hands of government, that is, we can't take them violently out of the hands of government, all we can do is by some sly roundabout way introduce something that they can't stop.
>
> -Friedrich Hayek, 1899-1992

Satoshi Nakamoto's answer to that riddle in 2008 was to avoid a centralized cluster and make a peer-to-peer money system based on a distributed ledger.

> I've been working on a new electronic cash system that's fully peer-to-peer, with no trusted third party.
>
> Governments are good at cutting off the heads of a centrally controlled networks like Napster, but pure P2P networks like Gnutella and Tor seem to be holding their own.
>
> -Satoshi Nakamoto, two separate quotes from 2008

Nakamoto's invention of bitcoin in 2008, which cited a number of projects, indeed became the first widely successful and credibly-decentralized internet money after it was launched in early 2009. In the genesis block, he referenced a topical newspaper headline about British bank bailouts, during the heart of the global financial crisis.

> " The Times 03/Jan/2009 Chancellor on brink of second bailout for banks
>
> -Bitcoin Genesis Block

The Bitcoin network is a distributed database, also known as a public ledger or "triple entry bookkeeping". It's a system that allows all participants around the world to come to a consensus on the state of the ledger every ten minutes on average. Because it's highly distributed and relatively small in terms of data, participants can store a full copy of it and reconcile it constantly with the rest of the network, with a specific protocol for determining the consensus state of the ledger. In addition to storing the whole database, participants can store their own private keys, which allow them to move coins (or fractional coins) around to different public addresses on the ledger.

If participants hold their own private keys, then their bitcoins represent assets

rather than currency, as long as other people recognize them as having value.

*Chart Source: LookIntoBitcoin.com*

After Nakamoto showed the way, there have been over fifteen thousand other cryptocurrencies created. Some of them are competitors to the bitcoin network, while others are smart contract platforms to serve other purposes.

any traction against the bitcoin network (with none of them sustaining above 5% of the network value of bitcoin), while a select few that aim to be used as smart contract utility tokens instead have retained rather large network valuations for longer periods of time.

Many people argue that bitcoin has achieved critical mass in terms of its network effect, security, immutability, and decentralization, such that while other digital assets may persist to fulfill other use-cases, none of them have a reasonable chance of competing with bitcoin in terms of being hard money. Fidelity published a good paper on this topic called Bitcoin First. Here's the summary:

> " In this paper we propose:
>
> -Bitcoin is best understood as a monetary good, and one of the primary investment theses for bitcoin is as the store of value asset in an increasingly digital world.
>
> -Bitcoin is fundamentally different from any other digital asset. No other digital asset is likely to improve upon bitcoin as a monetary good because bitcoin is the most (relative to other digital assets) secure, decentralized, sound digital money and any "improvement" will necessarily face tradeoffs.
>
> -There is not necessarily mutual exclusivity between the success of the Bitcoin network and all other digital asset networks. Rather, the rest of the digital asset ecosystem can fulfill different needs or solve other problems that bitcoin simply does not.
>
> -Other non-bitcoin projects should be evaluated from a different perspective than bitcoin.

> -Bitcoin should be considered an entry point for traditional allocators looking to gain exposure to digital assets.
>
> -Investors should hold two distinctly separate frameworks for considering investment in this digital asset ecosystem. The first framework examines the inclusion of bitcoin as an emerging monetary good, and the second considers the addition of other digital assets that exhibit venture capital-like properties.

Other blockchains that attempt to increase transaction throughput on the base layer or add more computational functionality on the base layer, generally sacrifice some degree of decentralization and security to do so. Blockchains that attempt to have more privacy on the base layer generally sacrifice some degree of supply auditability.

Bitcoin is updated slowly over time via optional soft forks, but the underlying foundation is maximized towards decentralization and hardness, more so than throughput or additional functionality. Layers built on top of it can increase throughput, privacy, and functionality.

The discovery of a credible way to maintain digital scarcity, and an invention of a peer-to-peer money based on that discovery, was in some ways inevitable, although the specific form that it first appeared in could have been designed in a number of ways. The foundations of the internet were put forth in the 1970s, and so was the concept of a Merkle tree. Throughout the 1980s and 1990s the internet as we know it came into being, as more and more of the world's computers were networked together. Proof-of-work using computer systems was invented in the 1990s, and the SHA-256 encryption was published in the early 2000s. Nakamoto put a bunch of these concepts together in a novel way in 2008, and had the right macroeconomic backdrop and right design decisions to have it succeed for well over a decade and counting.

### Bitcoin's Bottom-Up Monetization

If we go back to the gold standard for a moment, the key reason why paper claims were built on gold was to improve its medium-of-exchange capabilities. Ray Dalio described it well:

> "Because carrying a lot of metal money around is risky and inconvenient and creating credit is attractive to both lenders and borrowers, credible parties arise that put the hard money in a safe place and issue paper claims on it. These parties came to be known as "banks" though they initially included all sorts of institutions that people trusted, such as temples in China. Soon people treat these paper "claims on money" as if they are money itself.
>
> -Ray Dalio, The Changing World Order

Bitcoin on the other hand is a bearer asset that is safe to self-custody in large amounts and can be sent peer-to-peer around the world over the internet. Therefore, it removes the need for paper abstraction. Some holders will still prefer custodians to hold it for them, but it's not necessary like it is with large amounts of gold, and thus the units of the network are less prone to centralization. Unlike gold, bitcoin in large quantities is easy to transfer globally, and take custody of.

From the start, the bitcoin network was designed as a peer-to-peer network for the purpose of being a self-custodial medium of exchange. It's not the most *efficient* way to exchange value, but it's the most *unstoppable* way to do so online. It has no centralized third parties, no centralized attack surfaces, and sophisticated ways of running it can even get around rather hostile networks. Compared to altcoins, it is far harder to attack due to its bigger network effect and larger hash rate.

For example, one of the early use cases for bitcoin back in 2010/2011 was that Wikileaks was dropped by PayPal and other payment providers, so it began accepting bitcoin donations instead. Satoshi himself expressed concern about this on the forum at the time, due to bitcoin still being in its infancy back then relative to the amount of attention this would bring.

Kind of like how a tank is designed to get from point A to point B through resistance, but is not well-suited for commuting to work everyday, the base layer of the bitcoin network is designed to make global payments through resistance, but is not well-suited for buying coffee on the way to work.

In that sense, the bitcoin network has utility, for both ethical and unethical participants (just like any powerful technology). And because it is broken up into 21 million units (each with eight decimal places, resulting in 2.1 quadrillion sub-units), it is a finite digital commodity.

And that's how Satoshi described it:

> As a thought experiment, imagine there was a base metal as scarce as gold but with the following properties:
> – boring grey in colour
> – not a good conductor of electricity
> – not particularly strong, but not ductile or easily malleable either
> – not useful for any practical or ornamental purpose
>
> and one special, magical property:
> – can be transported over a communications channel
>
> If it somehow acquired any value at all for whatever reason, then anyone wanting to transfer wealth over a long distance could buy some, transmit it, and have the recipient sell it.

In addition to sending them online, bitcoins in the form of private keys can be physically brought with you globally. You can't bring a lot of physical cash or gold through an airport and across borders. Banks can block wire transfers out of their country, or even within the country. But if you have bitcoins, you can bring an unlimited amount of value globally, either on your phone, or on a USB stick, or stored elsewhere on some cloud drive you can access from anywhere, or simply by memorizing a twelve-word seed phrase (which is an indirect way of memorizing a private key). It's challenging for governments to prevent that without extremely draconian surveillance and control, especially for technically-savvy citizens.

This utility combined with an auditable and finite number of coins eventually attracted attention for its monetary properties, and so bitcoins acquired a monetary premium. When you hold bitcoins, especially in self-custody, what you are holding is the stored-up ability to perform global payments that are hard to block, and the stored-up ability to transfer your value globally if you want to. You are holding your slot on a global ledger, similar to holding valuable domain names, except that unlike domain names, bitcoins are decentralized, fungible, liquid, and self-custodial. It could be an insurance policy for yourself one day, or you could simply hold it because you recognize that capability to be valuable to others, and that you could sell that capability to someone else in the future.

Bitcoin is becoming a rather salable good, in other words. And with a higher stock-to-flow ratio than gold.

It's volatile, but that's in large part because it monetized from zero to a trillion-dollar market capitalization in twelve years. The market is exploring this technology and trying to determine its total addressable market as more and more people buy into it over time. It's an asset that is still only held by about 2% of the global population, and it's a tiny fraction of global financial assets.

Censorship-resistance is a significant feature when it comes to payments, and self-custodying money that cannot be diluted with more supply is a significant feature when it comes to savings.

To many people in developed countries, those features might not seem important, because we are privileged and take our freedom and comfort for granted. But for a large portion of the world, being able to bring self-custodied wealth with you if you have to leave your country is immeasurably valuable. When Jews fled Nazi-controlled Europe, they had trouble bringing any valuables with them. When people left the failing Soviet Union, they could only bring the equivalent of $100 USD with them. When people today want to leave Venezuela, Syria, Iran, Nigeria, China, eastern Ukraine, or any number of countries, they sometimes have a rough time bringing a lot of value with them, unless they have self-custodied bitcoins. Millions (and arguably billions) of people today can understand the value of this feature.

Reuters has documented Putin's domestic political opposition using bitcoins as Putin's establishment cuts them off from their banking relationships. The Guardian has documented Nigerians using bitcoins as they protested their government against police violence and had their bank accounts frozen. Chinese people have used it to transfer value through capital controls. Venezuelans have used it to escape hyperinflation and transfer value out of their failed state. One of the earliest use-cases for it was to pay Afghani girls with a type of money that their male relatives could not confiscate, and that they could bring with them out of the country when they leave. I'll dive more into these examples later. In 2022, Canada used emergency powers to freeze financial accounts of protestors, and people who donated to protestors, before charging them with any crimes.

The limited scalability of bitcoin's base layer has not been an issue so far, because there is only so much demand for tank-like censorship-resistant payments. And as development has continued since bitcoin's launch, the

Lightning network, for example, is a series of smart contracts that run on top of the bitcoin network base layer and allow for custodial or non-custodial rapid payments online or in person with a mobile phone, to the point where they can easily be used to buy coffee, and with practically no limitation on transactions per second. The Liquid network, as another example, is a side chain that wraps bitcoins into a federated network for rapid transfers, better privacy, and additional features as well, in exchange for some security trade-offs.

In that sense, bitcoins began as digital commodities that had utility value as an internet-native and censorship-resistant medium of exchange for people that need that capability. Bitcoins eventually acquired a monetary premium as an emergent and volatile store of value (an increasingly salable good), and began to be held more-so for their scarcity than for their medium-of-exchange capabilities. And then over time, the network developed additional ways to enhance the network's medium-of-exchange capabilities beyond their initial limitations.

Too many people look at bitcoin and say, "the base layer can't scale so that everyone in the world can make all of their transactions with it", but that's not the point of what it's for. The base layer is a censorship-resistant payments and settlement network with an auditable supply cap that has the capacity to handle hundreds of thousands of transactions per day, and layers built on top of it can be used for more frequent transactions than that if desired.

Kind of like how we don't use Fedwire transfers to buy coffee, bitcoin base layer transactions are not well-suited to buying coffee. Visa transactions that run on top of Fedwire, or lightning transactions that run on top of bitcoin, can be used to efficiently buy coffee. Or even custodial payment methods like Cash App and Strike that run on top of the bitcoin/lightning network can be used if censorship-resistance is not needed. The base layer of the bitcoin network is not competing with things like Visa; it is competing with central bank

root layer, built on computer networking technologies and internet protocols rather than channels between central banks and commercial banks.

It's also worth understanding Gresham's law, which proposes that "bad money drives out good". Given the choice between two currencies, most people spend the weaker one and hoard the stronger one. Bitcoin's low current usage as a medium of exchange is not a bug; it's a feature of a system with low supply issuance and a hard cap at 21 million units, especially in places where it is not legal tender and so every transaction is a taxable event. When a tank-like medium of exchange is needed, or for certain other niche use-cases, bitcoin is useful for its payment utility. Otherwise, it's most often held for its monetary premium as a scarcer asset than dollars and other fiat currencies, and represents the stored-up ability to perform tank-like payments in the future.

Bitcoin as a network and surrounding ecosystem went through multiple boom/bust cycles so far, and in each one, larger pools of capital became interested in it. In the first era, the user experience was challenging and required technical understanding, so it was mainly computer scientists and enthusiasts building and exploring the technology. In the second era, bitcoin became a bit easier to use and reached enough liquidity to have a quoted price in dollars and other fiat currencies, and so it became noticed by early speculators as well as dark net buyers/sellers. In the third era, it reached early mainstream adoption, in the sense that exchanges with proper security protocols could operate with bank connections, provide more liquidity to the market, and improve the user experience so that everyday people could more easily buy some. In the fourth era, institutional-grade custodians entered the market, which allowed pensions, insurance companies, hedge funds, family offices, and sovereign wealth funds to safely allocate capital to it.

It'll be interesting to watch how the bitcoin ecosystem develops going forward. Will it remain rather decentralized, or will it eventually become more

to maintain robust market share of the digital asset ecosystem, against thousands or tens of thousands of competitors that dilute each other and try to take some of bitcoin's monetary premium? I'm bullish and optimistic on the network but it's not without challenges and risks.

**Corporate Stablecoins**

Corporate custodial stablecoins were created via smart contracts to apply blockchain technology to fiat currency. With these systems, a custodian of dollars could issue tokens on a smart contract blockchain, and each of those tokens is redeemable 1-for-1 from the custodian for dollars.

To create custodial stablecoins, a client deposits dollars with the issuer, and is issued stablecoins in return. To redeem stablecoins, a client deposits stablecoins and is issued dollars in return. Different custodians have different track records for how securely they hold the collateral in dollars; users have to trust the custodian not to gamble away the funds on bad investments or fraud. Attestations and/or audits by third party accounting firms can add assurances about the safety of the collateral.

Once stablecoins are issued, people can then use whichever blockchain they are issued on to send and receive stablecoin payments between themselves, peer-to-peer, with no additional centralized third party. From a user perspective, stablecoins are a significant technological leap over existing bank payment systems, especially for international payments, or large domestic payments. You can send someone on another continent a million dollars at 2am on a Sunday night and they can receive it in minutes, and both sides can verify the transaction on the blockchain in real time.

They will naturally face ongoing government regulation and be controlled and surveilled as part of the banking system in many cases, but it's clear that they have utility for actual payments and will probably get increasingly

incorporated into financial systems, either in the form of central bank digital currencies or private-but-highly-regulated stablecoin issuers.

This is simply due to automation and superior technology. When you send a wire transfer, the bank has to *actively do something* to process that transaction. And wires often get delayed or blocked or run into other problems as they flow between banks. From the users' perspective, it's often unclear which bank it got stuck in or who to call, and thus it sometimes takes days to resolve. With stablecoins, it's the opposite. The automatic nature of the blockchain allows for peer-to-peer transactions handled by software, including internationally and including with large amounts of money. The custodians are *passive* in that regard and let the technology work for them, and only act in the event that they want to blacklist some of their tokens for some reason that they detected.

Anyone who does a lot of international wire transfers, and then has used stablecoins, will generally say that stablecoins are way better to use.

In other words, regulated stablecoins allow for an automated peer-to-peer international payment system, but that has an overlay of control based on know-your-customer and anti-money-laundering "KYC AML" laws, as well as a significant element of custodial trust.

**Central Bank Digital Currencies**

Some countries want to take the stablecoin concept further, and completely nationalize it within their jurisdiction. This uses similar technology to stablecoins but doesn't need a blockchain, because it's not decentralized.

Starting with China studying and learning from bitcoin and stablecoins for over five years now, these technologies are now being used to create central bank digital currencies. These are central-bank issued fiat currencies that are

digitally-native, able to operate over the internet rather than going through the historic global banking system "pipes".

From the government perspective, the usefulness of a pure central bank digital currency is that the government can:

1. send international payments without the SWIFT system
2. try to give banking access to the non-banked or under-banked populations
3. track and surveil any transaction, including with Big Data/AI technologies
4. blacklist or block certain transactions that violate their rules
5. add expiration dates or jurisdiction limitations to currency
6. take away money from citizen wallets for various violations
7. give money to citizen wallets for stimulus or rewards
8. impose deeply negative interest rates on citizen account balances
9. program money to have different rules for different groups
10. reduce the control and fee pressure that commercial banks have over the system

In other words, a central bank digital currency can be more efficient, cheaper, and easier to use than many existing payment systems. However, in such a system, your currency can also be surveilled, given, taken, and/or programmed by the issuer to only work in authorized situations.

Agustin Carstens, head of the Switzerland-based Bank for International Settlements (basically the central bank of central banks) had an interesting quote on CBDCs last year:

> " For our analysis on CBDC in particular for general use, we tend to establish the equivalence with cash, and there is a huge difference there. For example in cash, we don't know for example who is using

> thousand peso bill today. A key difference with a CBDC is that central bank will have absolute control on the rules and regulations that determine the use of that expression of central bank liability. And also, we will have the technology to enforce that. Those two issues are extremely important, and that makes a huge difference with respect to what cash is.

### A Spectrum of Control

From the summaries of the sections above, there are multiple types of digital assets. There are decentralized bearer assets like bitcoins, and there are digital representations of fiat currency like corporate stablecoins and central bank digital currencies. There are also other private blockchain monies, such as utility tokens or gaming tokens.

Some digital assets, like bitcoins, *reduce* the government's ability to interfere with your money, since you can self-custody it and send it to whoever you want. As the Guardian covered back in July, when Nigerans began protesting their government for police violence, and found their bank accounts frozen for doing so, many of them turned to using bitcoins to remain operational.

> "Last October, Nigeria was rocked by the largest protests in decades, as many thousands marched against police brutality, and the infamous Sars police unit. The "EndSars" protests saw abuses by security forces, who beat demonstrators, and used water cannon and teargas on them. More than 50 protesters were killed, at least 12 of them shot dead at the Lekki tollgate in Lagos on 20 October
>
> The clampdown was financial too. Civil society organisations, protest groups and individuals in favour of the demonstrations who

> were raising funds to free protesters or supply demonstrators with first aid and food had their bank accounts suddenly suspended.
>
> Feminist Coalition, a collective of 13 young women founded during the demonstrations, came to national attention as they raised funds for protest groups and supported demonstration efforts. When the women's accounts were also suspended, the group began taking bitcoin donations, eventually raising $150,000 for its fighting fund through cryptocurrency.

And many merchants, facing sanctions, used bitcoins to trade internationally (also from the Guardian article):

> "His business – importing woven shoes from Guangzhou, China, to sell in the northern city of Kano and his home state of Abia, further south – had been suffering along with the country's economy. The ban threatened to tip it over the edge. "It was a serious crisis: I had to act fast," Awa says.
>
> He turned to his younger brother, Osy, who had begun trading bitcoins. "He was just accumulating, accumulating crypto, saying that at some point years down the line it could be a great investment. When the forex ban happened, he showed me how much I needed it, too. I could pay my suppliers in bitcoins if they accepted – and they did."

Similarly, Reuters has been reporting on a number of occasions that Russian opposition leader and anti-corruption lawyer Alexei Navalny uses bitcoins in his organization to get around government blockades:

> "

> founded which conducts investigations into official corruption.
>
> "They are always trying to close down our bank accounts – but we always find some kind of workaround," said Volkov.
>
> "We use bitcoin because it's a good legal means of payment. The fact that we have bitcoin payments as an alternative helps to defend us from the Russian authorities. They see if they close down other more traditional channels, we will still have bitcoin. It's like insurance."

One of the most touching stories was reported by Reuters as well. In the early years of bitcoin, an Afghan woman paid many girls with bitcoins, because they were otherwise unbanked and their male relatives would often try to steal from them, since they didn't necessarily have much of a right to their own property. The self-custodied aspects of bitcoin then allowed many of the girls over the years to leave the country with their funds, which would be impossible with most other assets:

> "When Roya Mahboob began paying her staff and freelancers in Afghanistan in bitcoin nearly 10 years ago, little did she know that for some of these women the digital currency would be their ticket out of the country after the fall of Kabul in August.
>
> Mahboob, a founder of the non-profit Digital Citizen Fund along with her sister, taught thousands of girls and women basic computer skills in their centres in Herat and Kabul. Women also wrote blogs and made videos for which they were paid in cash.
>
> Most girls and women did not have a bank account because they were not allowed to, or because they lacked the documentation for

Alex Gladstein has a massive archive of articles reporting on the various emerging market use-cases for bitcoin over the past several years, ranging from Sudan to Palestine to Cuba to Iran to Venezuela and more.

Anita Posch also has a great interview series called Bitcoin in Africa that explores these use-cases in that region. Bitcoin is a tool that tech-savvy people often use as defense against either double digit currency inflation or authoritarian financial system control.

We're even seeing this topic pop up in developed markets. Truckers in Canada protested the government and occupied and disrupted the capitol, and received donations from supporters on crowdfunding sites. Those crowdfunding sites ended up freezing and reversing the payments, so many of the participants turned to bitcoin as peer-to-peer money. The government then invoked the 1988 Emergencies act to freeze bank accounts of certain protestors and donors, and to try to blacklist certain bitcoin addresses to being brought to exchanges for conversion back into Canadian dollars.

People may agree or disagree with aspects of those protests but the pragmatic point about money in this context is, those who had their money entirely in banks were indeed frozen. Those who self-custodied their own digital assets, such as bitcoins, had certain conveniences removed from them but could still hold, move, and transfer their money in various ways.

In the broadest sense applying internationally (especially for developing markets with weaker rule of law where the majority of people live), I described the issue here:

Other digital assets, like CBDCs, are the *opposite* of this type of asset, and give the government more ability to surveil and censor your money, and in reality, it's not even your money. It's a liability of your country's central bank, and as Carstens eloquently articulated, each central bank wants to be able to determine how you can use their liabilities. The full ramifications of that statement can mean very different things depending on whether you live in a place like Norway or a place like China.

The European Central Bank published a working paper on CBDCs in early 2020 called "Tiered CBDC and the Financial System" where they outlined the ability of CBDCs to better control illicit payments and to allow for deeper negative interest rate policy, especially if physical banknotes are removed from circulation. This effectively corrals public savings into a digital money that the government can more easily debase and control as desired and gain more seigniorage from:

*Source: Tiered CBDC and the Financial System, ECB, January 2020*

This becomes particularly relevant when we consider that the government can always try to broaden its scope of what is "illicit", particularly in regards to protests and things of that nature. Basically, we have to ask ourselves not what the current political leadership would do with this technology, but also what all future political leadership who we don't know yet would do with this technology. What would Norway do with this technology, compared to what China would do with this technology?

Although bitcoin has thus far been somewhat more appreciated by libertarian and fiscally conservative people on average, this feature is why there are also some progressive/left voices out there that identify bitcoin as a tool for their goals as well. At the end of the day, bitcoin is more of an anti-authoritarian monetary technology than it is a "left" or "right" monetary technology. The Human Rights Foundation in particular has made extensive use of it for their international activities.

Critics of bitcoin often leave these humanitarian or anti-authoritarian use-cases out (or don't even realize them), and instead refer to bitcoin as being primarily used to buy drugs or ransomware or money laundering, which is a really outdated (or deliberately misleading) view at this point. Firms such as Chainalysis that perform blockchain analysis for law enforcement and other clients have found that bitcoin and overall cryptocurrency usage for illicit activities involves less than 1.5% of bitcoin/crypto transaction volume over the past several years, which is less than the percentage of fiat transactions used for illicit activities.

Bitcoin went through an early phase in 2011 through 2013 where it was used for online drug purchases and such, until authorities responded with a crackdown on that usage by going after the centralized marketplaces that enabled it. Just like how the invention of the pager was used by both drug dealers and doctors in the 1970s and 1980s, bitcoin has gone through phases where criminals used it and humanitarians used it for their purposes. Both of those groups in particular have an incentive to quickly adopt to new technologies to stay ahead of their state-sponsored competition, and it's important for western media to keep in mind that "illegal activities" in some countries includes protesting the government and other forms of free speech and expression and political opposition.

Like any powerful technology, bitcoin can be used for good or ill. As proponents of the technology like to say, bitcoin is "money for enemies"

hard to block payments for anyone. It's like a commodity; something that can be partially regulated within certain jurisdictions but that in the holistic sense, exists outside of anyone's control.

If we take a step back, we can catalogue the history of financial surveillance. For most of human history, financial transactions were rather private from the perspective of the government, because transactions mainly involved handing over physical money, which is hard to track. With the invention of modern banking, and then especially modern computer databases and electronic payments, transactions could be more easily tracked and surveilled. The Bank Secrecy Act of 1970 required financial institutions to report transactions over $10k USD to the government, which back then was the equivalent of about $75k USD in today's dollars. They never raised the threshold despite five decades of inflation, so over time without further laws being passed, their surveillance reporting requirements became applicable to smaller and smaller transactions.

When people use banks to send or receive money, it is easy for governments to impose restrictions on what sort of payments are allowed, which banks can then enforce. And some governments can even block other foreign governments from using the primary existing international payment methods. Bitcoin threatens that surveillance and control model because it empowers peer-to-peer transactions. The bitcoin network consists of people using free open source software to update a public ledger between themselves. It's basically just a sophisticated way of updating the equivalent of a distributed Google Spreadsheet with each other, without a centralized server. Governments trying to ban people from doing that is tantamount to banning the spread of information, and is therefore a lot harder to do than telling banks to report or block certain types of transactions.

Governments will be challenged by this technology, and many of them have, and will, push back against it. They can allocate law enforcement resources to

involved with serious crime), but will likely have trouble trying to retain control over benign transactions. They can use on-chain analysis to try to track transactions, they can enforce surveillance checkpoints around cryptocurrency exchanges, they can block banks in their jurisdiction from interfacing with any cryptocurrency exchanges, and at the extreme end they can put draconian punishments on people for using open source software to update a public ledger between each other. Meanwhile, developers continue to find ways to make the bitcoin network more private and to route around some of the challenges that can be put in its way. There are also some privacy-specific coins that people can resort to as well.

One way or another, these various types of digital money or currency are clearly in our future in some form or another. Depending on where we live and choices we make, we are more likely to experience some than others, ranging from bitcoins to corporate stablecoins to central bank digital currencies.

## Proof-of-Something

A topic popularized by bitcoin is the term “proof-of-work”.

The concept was invented in various ways by cryptographers in the late 1990s, including notably by Dr. Adam Back in the form of “Hashcash”- a money-like mechanism to reduce email spam and denial-of-service attacks by making them have a small computational cost.

Satoshi Nakamoto’s bitcoin white paper referenced Back’s work, and used proof-of-work as one of its core aspects.

Nowadays, various digital assets have expanded on this concept in the form of “proof-of-stake”, “proof-of-history”, “proof-of-transfer”, “proof-of-burn”, “proof-of-space” and so forth. There are multiple attempts at maintaining

In any form, money is proof of *something*. This section explores three popular examples of proof-of-work, proof-of-stake, and proof-of-force.

Proof-of-work assets are created or harvested from mining activities. Proof-of-stake assets are created by breaking a project into pieces and selling some of those pieces to others. Proof-of-force assets, or fiat currency, is created by governments and their designated commercial parties (holders of banking licenses).

**Proof-of-Work**

When we go back and look at the example of rai stones, they were well-understood among their users to be a powerful proof-of-work mechanism. In addition to having a high stock-to-flow ratio until modern technology interfered with that, each stone is an undeniable proof that a massive amount of work occurred to create it and put it where it is.

A team of young men had to travel hundreds of miles to another island, quarry for the stone with ancient tools, bring a multi-ton stone block back on their wooden boats, and then carve it and move it into place on their home island. The amount of work that was required to do this is what limited the flow (new annual supply), and maintained the high stock to flow ratio for a long time. The bigger the stone, the more work it took to produce it and get it there.

Gold, of course, is historically the best example of proof-of-work, and it has stood the test of time unlike anything else. After painstakingly searching for gold deposits, it takes a tremendous amount of mechanical effort to move tons of earth for grams of gold, and then it has to be refined into its pure form. Each gold coin or gold bar represents literally tons of rock moved and sorted through, and gold resists degradation better than other elements. The Earth's crust consists of less than 0.0000004% gold, compared to over 28%

improves and we get better at finding and retrieving gold, we run out of the easiest deposits, and so it keeps getting harder, which offsets our improving technology.

Basically, proof-of-work is just that: proof that work was done. Since work is inherently scarce, we tend to recognize proof-of-work as being evidence of value, *but only if the finished good in question has properties of money*. And that's an important distinction; we don't pay for non-monetary goods or services based on how much work went into them; we pay for them based on how much utility they provide to us.

In other words, something akin to the Labor Theory of Value doesn't apply to utility goods, but does apply to monetary goods.

This is because market participants will naturally try to arbitrage any good that acquires a monetary premium above what it offers in terms of its utility value. Monetary goods that don't require work inevitably get reproduced and devalued (thus leaving only those that do require work as proper monies), whereas goods with no monetary premium are not worth reproducing endlessly. Basically, when it comes to money, a large amount of work to produce a unit, and a persistently high stock-to-flow ratio, are essentially the same thing. That work requirement is what keeps a commodity's stock-to-flow ratio high, and any commodity that can't maintain a high stock-to-flow ratio in the face of ever-advancing technology eventually fails as money. Only the scarcest of monies can maintain a persistent monetary premium over its utility value, because that monetary premium continually invites attempts at debasement.

For bitcoin, a new block of transactions is produced every ten minutes on average, and contains a cryptographic hash of the block before it, which connects the blocks to form a chain. It takes work (computer processing power) to solve that puzzle and find the new block that fits. The blockchain

that a large amount of work was done. Copies of the blockchain are distributed and continually updated across tens or hundreds of thousands of computers worldwide.

A transaction recognized by the chain becomes essentially unchangeable, as it is buried under thousands of hashed blocks and widely distributed on those global computers.

And because bitcoin has a much larger network effect than most other cryptocurrencies, it is far more costly to attack the network than it is to attack most other cryptocurrencies. This, along with the fact that the node network is sufficiently decentralized and the monetary policy (or more accurately, initial coin distribution policy) can't realistically be changed, is what has made bitcoin able to accumulate a persistent hard money premium that other cryptocurrencies have had trouble maintaining. However, in the grand scheme of things, it's still only thirteen years old.

I have a longform research piece on bitcoin's proof of work and energy consumption here.

**Proof-of-Stake**

Proof-of-stake is an equity-like system whereby holders of an asset determine how that asset functions. In other words, each coin can serve as a vote for the network.

Much like proof-of-work, we can translate it back into analog examples. In particular, proof-of-stake is commonly used in corporate ownership. The larger the number of shares of a company you own, the more say you have in terms of electing board members to run the company, and supporting or denying shareholder proposals. If you, or a group of entities that follow you, can control 51% of the shares, you effectively control the entire company.

Similarly, some blockchains have used this approach. Rather than mining for coins with real-world resources, users create new coins by signing transactions as a validator. In order to be a validator, users have to prove that they have a certain number of coins. Some of the pre-bitcoin attempts at digital money used strategies like this, and many of the post-bitcoin attempts use it in blockchain form.

However, unlike corporations, proof-of-stake blockchains require circular logic. Corporations use an external entity (a transfer agent and registrar) to keep track of who owns each share. In proof-of-stake blockchains, it's like a corporation acting as its own transfer agent and registrar; the coinholders determine the state of the ledger, and the ledger is what says who the coinholders are.

Therefore, proof-of-stake systems need to be "always on" to function, and are highly complex. They have no inherent disaster recovery potential if the blockchain goes offline, because making alternative copies of the blockchain has no cost, and there's no way to determine the "real" blockchain other than via agreement of major parties (a.k.a. a form of governance) if it is recovering from that offline state.

In contrast, proof-of-work systems are ledgers with decentralized and automated transfer agents and registrars. The coinholders do not determine the state of the ledger, the miners do, via energy expenditure. A proof-of-work system is not based on circular logic; even if the entire blockchain goes offline, it can be restarted because the longest chain can be identified and continued.

In other words, a proof-of-work blockchain is like non-volatile memory and a proof-of-stake blockchain is like volatile memory.

Another risk with proof-of-stake systems in both the analog and digital world is that they tend to centralize over time into an oligopoly. Since it doesn't

time, wealth tends to compound into more wealth, which they can then use to influence the system to give themselves even more wealth, and so on.

Adam Back described this succinctly a while ago:

> “ You see that with other commodity money, like physical gold. It’s a system that works because money has a cost. I think money that doesn’t have a cost ultimately ends up being political in nature. So people closer to the money, the so-called Cantillon Effect, are going to be advantaged.

In digital systems specifically, another challenge is that proof-of-stake as a consensus model is a lot more complex than proof-of-work and prone to more attack surfaces. If a proof-of-stake chain gets split or maliciously copied, it’s not self-evident which chain is the real one, and it becomes a human/political decision among oligopolistic participants to canonize a chain. However, in a proof-of-work system, the real chain is instantly verifiable, because by definition the chain that follows the node-consensus ruleset and that has more work is the real one.

In other words, what makes proof-of-stake blockchains inherently equity-like is that they require some form of ongoing governance, whereas proof-of-work blockchains (especially ones decentralized enough that they can’t really change their monetary policies) are more commodity-like. These differences can be benefits or drawbacks depending on what participants want out of the system. The collective existence of both digital commodities and digital equities in my view represents a novel new era for asset classes, and we’ll see where they may be successfully applied.

I have a longform research piece that includes an overview of proof-of-stake here.

As described by Warren Mosler, a founder of the MMT school of economic thought mentioned earlier in this article, fiat currency is basically proof-of-force, which is why it can win out over proof-of-work money for long stretches of time.

Demand for government paper (or digital equivalents) is created by the government's taxes on the population, which can only be paid in units of that paper. Failure to pay taxes results in losing assets, going to jail, or if resisting those prior consequences, getting shot by police. Proof-of-force systems convince or coerce people in their jurisdiction to use a softer/devaluing money, by placing taxes, frictions, and other obstacles on any money that is harder than their own, or in some cases outright banning competing monies by making it a felony to use them.

Of course, proof-of-force has existed for thousands of years, prior to the invention of fiat currency. Any warlord, kingdom, or empire that demanded some tribute from the peoples of the land it ruled were familiar with the concept of proof-of-force. The purpose could be for malevolent ends, or it could be for benevolent purposes to provide order for society, and collect some percentage of resources into the common good. Even democracies use proof-of-force as an organizational method. Nature abhors a vacuum, and humans consistently congregate into hierarchies and societal structures. In other words, not every political leader was like Caligula; some of them were more like Marcus Aurelius, or were democratically elected.

In most eras, that tribute took the form of commodity monies, such as gold or other loot that was already recognized as money via proof-of-work. However, in the modern era, governments have eliminated the proof-of-work component from the equation via technology (banking systems and efficient nationwide communication systems) and so when we think of the dollar, the euro, the yen, and other fiat currencies, they basically represent just proof-of-force. When we say that the dollar is "backed up by the full faith and credit of

by the ability of that government to collect taxes by any means necessary including force (and backed up by the petrodollar system; the ability of the US government to maintain a currency monopoly on energy pricing worldwide).

That sounds like hyperbole, so we can put it in context and dial it back a bit. Even Switzerland, well-known for its hundreds of years of geopolitical neutrality in the face of war, inherently uses proof-of-force to collect taxes in its fiat currency. So, even the most benign and nonviolent society, for the least belligerent purposes possible, still uses this proof-of-force mechanism to ensure the societal usage of its government-issued money, as a way to provision the government. In benign environments, force is sharply minimized by the fact that people vote for the government, or can leave the country and renounce citizenship if they do not wish to play by these rules, and thus can choose another country's ruleset if that other country will let them in.

To put it bluntly, if you don't pay your taxes, *and in a form of legal tender accepted by the government*, you eventually get a knock on your door from people with guns, and/or you'll have to leave and go somewhere else. That remains the case unless or until the country's legal tender breaks down enough that the majority of people can't/won't use it and the government is unable to enforce its use with that level of currency rebellion, which happens during hyperinflations and near-hyperinflations, including in many developing countries in modern times.

With a stock-to-flow ratio averaging somewhere between 5x to 20x in most cases, major fiat currencies have higher stock-to-flow ratios than most commodities, but lower stock-to-flow ratios than bitcoin and gold. However, in addition to having a moderately high stock-to-flow ratio, fiat currency benefits from the unique backing by the government, including active stabilization to try to reduce volatility, which is what gives it a degree of staying power.

I have a longform piece that describes the process of how fiat currency is created and destroyed here.

## Final Thoughts: Think Outside of the Box

When money changes in a society, it always feels weird for people who go through it.

Imagine being someone who used shells for money their whole life, like your mother and your grandmother and your great-grandmother before you. And then, due to interactions with a foreign people, shiny yellow and gray metal circles with pictures of faces on them are starting to be used as money instead and seem to be displacing your shells. The foreigners, with better technology, can seemingly produce all the shells they want (which devalues them), but their shiny metal circles are harder to make and thus seem invulnerable to devaluation.

Or imagine using gold and silver coins as money your whole life, like your father and your grandfather and your great-grandfather, after thousands of years of global history of these things being used as money. And then, due to changing technology and government mandates, you're supposed to use pieces of paper that are backed by gold instead and treat them the same way, and it's illegal to actually own gold. And then, they take away the peg to gold and you're still supposed to keep using these papers for the same value anyway, even as the quantity of these papers seems to keep increasing. The successful papers, being actively managed, tend to be rather stable most of the time even though they degrade in value over time.

And lastly, imagine using these unbacked papers as money your whole life. The interest rates on those papers at first are higher than price inflation and they're rather stable in terms of purchasing power from year to year, but over time the interest rates keep going down until they are well below the

holding those papers. And then some anonymous entity comes along and creates internet money that works via encryption and algorithms that you don't fully understand, but it seems to keep growing in users and value compared to other assets for over a decade. Nobody can make more than the pre-programmed amount of it, it can be used for peer-to-peer domestic or international payments, and it can be self-custodied and transferred more easily and more securely than any prior money. But then we have questions about its technical risks, questions about whether governments can successfully prevent it from spreading, questions about its volatility, and other challenges along those lines that could cause it to stagnate or fail.

What do we do in these situations?

Well, I think the first rational thing is to be skeptical. We can't just dive all-in to anything new that people claim is money.

In fact, honestly at first we can probably ignore it, since the probability of any given new thing becoming money is low. It's pretty rare in human history that a serious new form of money emerges. But then if it doesn't go away, and indeed keeps surviving from multiple 80%+ drawdowns over more than a decade to greater and greater heights of increasing monetization, then realistically we need to research it, test its hardness, and envision all the ways it could conceivably fail.

If we happen to have expertise or interest in that field for one reason or another we might jump onto it quicker, or we might go about our lives and let it continue growing, so that we learn more about it and then maybe buy a little bit and get to understand it. If competitors spring up, we probably should investigate some of those as well, and watch how they behave, and understand the differences. And then over time, we can mostly let the market answer our questions for us. We can hold an amount of this new money that makes sense for our risk profile, and let it appreciate (or not) over time.

If it does not appreciate, then that answers a set of questions, and we risked very little. If it does appreciate, then we continue to watch this asset gain a larger and larger monetary premium. It then generally becomes owned by more people and becomes a larger percentage of what we own because it grows in value faster than our other assets. Due to Gresham's law, it won't be readily spent too often, and will instead have a tendency to be hoarded, and only spent when necessary or in niche circumstances for its tank-like payment properties. The vast majority of participants will treat it as a long-term financial asset. It if becomes very large and dominant and its volatility goes down over time, its usage in spending will likely go up.

Looking out over the long run, it's clear that money will become increasingly digital. The question is, will stateless peer-to-peer bearer assets like bitcoin become a persistently important version of money, worth trillions of dollars in market capitalization, or will state-created CBDCs or state-regulated corporate stablecoins be the main path forward instead? And to the extent that they coexist with each other, how much market share can we expect each one to take? That's a topic I'll continue to analyze over time.

As I close out this article, I'll circle back to an earlier example of bitcoins being used as confiscation-resistant self-custodied payment for Afghani women and girls nearly a decade ago. Alex Gladstein documented what became of some of them:

> A few of the women did keep their bitcoin from 2013. One of them was Laleh Farzan. Mahboob told me that Farzan worked for her as a network manager, and in her time at Citadel Software earned 2.5 BTC. At today's exchange rate, Farzan's earnings would now be worth *more than 100 times* the average Afghan annual income.
>
> In 2016, Farzan received threats from the Taliban and other conservatives in Afghanistan because of her work with computers

her family and selling their home and assets to pay brokers to take them on the treacherous road to Europe.

Like thousands of other Afghan refugees, Farzan and her family traveled by foot, car and train thousands of miles through Iran and Turkey, finally making it to Germany in 2017. Along the way, dishonest middlemen and common thieves stole everything they brought with them, including their jewelry and cash. At one point, their boat crashed, and more belongings sank to the bottom of the Mediterranean. It's a tragic story familiar to so many refugees. But in this case, something was different. Through it all, Farzan was able to keep her bitcoin, because she hid the seed to her bitcoin wallet on a piece of tiny, innocuous-looking paper. Thieves could not take what they could not find.

That's an example of bitcoin transporting value across borders in a circumstance where gold and cash would have failed. It can be done through a mobile phone, USB stick, piece of paper, cloud storage, or even just by memorizing a twelve-word seed phrase.

Whether the bitcoin network ultimately succeeds or fails in the long run, this global distributed ledger backed up by proof-of-work is clearly a form of money, and one that is worth understanding.

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